eLife

Reviewed Preprint
v1 • December 27, 2024
Not revised

Neuroscience

Acetylcholine modulates prefrontal outcome coding during threat learning under uncertainty

Gaqi Tu, Peiying Wen, Adel Halawa, Kaori Takehara-Nishiuchi

Department of Psychology, University of Toronto, Toronto, Canada • Collaborative Program in Neuroscience, University of Toronto, Toronto, Canada • Human Biology Program, University of Toronto, Toronto, Canada • Department of Cell and Systems Biology, University of Toronto, Toronto, Canada

eLife Assessment

This is an **important** study using a combination of optogenetics and calcium imaging to provide insight into the function of the cholinergic input to the prelimbic cortex in probabilistic spatial learning as it relates to threat. These data are timely in contributing to an ongoing discussion in the field about the role of phasic cholinergic signaling to the cortex, about which relatively little is known. The strength of the evidence is **incomplete** and could be improved by changes in task design and analyses, cross-validation of the conditions in calcium imaging, as well as the incorporation of control experiments to more definitively show it is indeed acetylcholine working in this circuit.

https://doi.org/10.7554/eLife.102986.1.sa4

Abstract

Outcomes can vary even when choices are repeated. Such ambiguity necessitates adjusting how much to learn from each outcome by tracking its variability. The medial prefrontal cortex (mPFC) has been reported to signal the expected outcome and its discrepancy from the actual outcome (prediction error), two variables essential for controlling the learning rate. However, the source of signals that shape these coding properties remains unknown. Here, we investigated the contribution of cholinergic projections from the basal forebrain because they carry precisely timed signals about outcomes. One-photon calcium imaging revealed that as mice learned different probabilities of threat occurrence on two paths, some mPFC cells responded to threats on one of the paths, while other cells gained responses to threat omission. These threat- and omission-evoked responses were scaled to the unexpectedness of outcomes, some exhibiting a reversal in response direction when encountering surprising threats as opposed to surprising omissions. This selectivity for signed prediction errors was enhanced by optogenetic stimulation of local cholinergic terminals during threats. The enhanced threat-evoked cholinergic signals also made mice erroneously abandon the correct choice after a single threat that violated expectations, thereby decoupling their path choice from the history of threat occurrence on each path. Thus, acetylcholine modulates the

encoding of surprising outcomes in the mPFC to control how much they dictate future decisions.

Introduction

In daily life, we rely on environmental cues to choose the course of action that would maximize positive outcomes and minimize adverse outcomes. This process requires tracking the discrepancies between received and expected outcomes, known as prediction errors(1–4). Prediction errors control how much a given outcome changes the expectation of subsequent outcomes, such that the larger the prediction error, the greater the change. This process, however, becomes complicated when outcomes follow cues or actions unreliably(5–7). Under such noisy conditions, even correct cues and actions occasionally result in large prediction errors. Such anomalies may indicate the need to accelerate learning to cope with a sudden, drastic environmental change. Alternatively, they may fall within a typically expected range of outcome variability and should be dismissed.

These two conflicting interpretations of surprising outcomes lead to "unexpected" and "expected" uncertainty, respectively(5–7). Humans and rodents can use both types of uncertainty to adjust the learning rate properly, and this essential ability relies on neuromodulatory systems, such as serotonin(8–13), noradrenaline(14–21), dopamine(22–26), and acetylcholine (ACh)(20,27–31). Parallel evidence also suggests that adaptive learning and decision-making under uncertainty rely on the integrity of the anterior cingulate cortex (ACC) in humans and monkeys(7,32–35) and its homolog, the medial prefrontal cortex (mPFC), in rodents(36–39). Neurons in these regions track outcome expectations by drastically changing the activity patterns depending on how reliably outcomes follow cues(40–44) or actions(45–48). They also signal prediction errors by scaling the magnitude of responses to outcomes according to how surprising it was(41,49–54). However, how these two systems interact to enable the proper handling of ambiguous outcomes remains unknown.

Here, we investigated the potential role of ACh in modulating prefrontal neural ensemble dynamics during adaptive learning and decision-making in a stable but probabilistic environment. We focused on ACh because (i) theories implicate ACh in computations of expected uncertainty(5,55), (ii) the mPFC receives extensive axonal projections of cholinergic cells in the basal forebrain (BF)(56–58), and (iii) these cholinergic cells emit phasic responses precisely time-locked to reward and punishment(58–65). Here, combining *in vivo* one-photon calcium imaging of mPFC cells with optogenetic manipulations of local cholinergic terminals in mice, we have shown that threat-evoked phasic cholinergic signals control the content of prediction error coding in the mPFC and limit the learning from surprising threats that are within an expected range of variability.

Results

Phasic cholinergic signals modulate the development of adaptive threat expectations

To investigate how threat-evoked phasic cholinergic signals modulate adaptive threat learning and its neural correlate in the mPFC, we designed a probabilistic spatial learning task in which mice chose one of two paths associated with different probabilities of threats to obtain rewards. Hungry mice went back and forth between two reward sites at the opposite corners of a square-shaped maze (**Fig 1A**). They always received sugar pellets at the reward sites [*P(Reward)* = 100%]; however, they also occasionally received an air puff toward the face/body in the middle of two paths connecting the reward sites (threat sites). On one of the paths (high-threat path, ht-p), an air puff was delivered 75% of the time [*P(Threat|ht-p)* = 75%], whereas on the other path (low-threat path, lt-p), it was delivered 25% of the time [*P(Threat|lt-p)* = 25%]. The mice received air puffs at the same probability on a given path regardless of which direction they ran. The timing of reward and threat delivery was controlled based on real-time tracking of mouse' position.

During the probabilistic spatial learning task, we conducted cellular-resolution calcium imaging from the prelimbic region (PL) of the mPFC with simultaneous optogenetic manipulations of the BF cholinergic terminals. We stimulated, rather than inhibited, cholinergic terminals for two reasons. First, because the relationships between ACh level and behavior follow an inverted U-shape, in which abnormally high ACh is as detrimental as abnormally low ACh(66,67). Second, we have shown that the stimulation of threat-evoked phasic cholinergic signals in the PL prevents mice from associating the threat with a preceding cue, while their inhibition facilitates association formation(58). Thus, the stimulation, but not inhibition, of threat-evoked cholinergic signals disrupts prefrontal neural processing during threat learning.

To limit the expression of excitatory opsins to cholinergic cells, the viral-mediated gene transfer approach was applied to mice expressing Cre recombinase in cholinergic cells ($ChAT_{(IRES)}$-Cre mice; **Fig 1B**). The mice were injected with conditional adeno-associated viral vectors (AAVs) for expressing a red-shifted excitatory opsin, ChrimsonR, with a florescent reporter (ChrimsonR group, n = 6) or the reporter alone (tdTomato group, n = 4) in the horizontal diagonal band of Broca (HDB) region of the BF (**Figs 1B**, S1A and S1B; based on our previous anatomical tracing result(58)). All mice were also injected with AAVs carrying a calcium indicator (GCaMP6f) under the control of a CaMKII promotor in the PL. Histological validation confirmed that the expression of ChrimsonR overlapped with cholinergic cells expressing ChAT, and their axons were located near putative excitatory cells expressing GCaMP6f in the PL (**Fig 1C**).

After the recovery period, all mice underwent fifteen days of training, divided into three stages (Fig S1C), pre-training [T1-5; *P(Threat|ht-p)* = 0%, *P(Threat|lt-p)* = 0%], probabilistic [P1-5; *P(Threat|ht-p)* = 75%, *P(Threat|lt-p)* = 25%], and random stages [R1-5; *P(Threat|ht-p)* = 50%, *P(Threat|lt-p)* = 50%]. During the probabilistic and random stages, we stimulated local cholinergic terminals unilaterally by delivering orange LED stimulations (300 msec, 20Hz, 8 mW/mm^2) to the imaged window. The LED stimulation was applied during ~50% of the air-puff delivery on both paths, allowing for monitoring threat-evoked activity with and without the LED stimulation (Fig S1D). During the pre-training stage (T5), both groups chose two paths with similar likelihood (t-test, $t(8) = 1.497$, $p = 0.173$; **Fig 1D**). When air-puff delivery and LED stimulation commenced (P1-5), tdTomato-expressing mice learned to choose the low-threat path preferentially (two-way mixed ANOVA, Session×Group interaction, $F(4,32) = 3.347$, $p = 0.021$; follow-up one-way repeated measure ANOVA, tdTomato, $F(4,12) = 12.469$, $p < 0.001$; **Fig 1D**). Initially, they approached both threat sites at comparable speeds (Fig S2A). After learning, they ran more slowly on the high-threat path than the low-threat path (three-way mixed ANOVA, Group×Path interaction, $F(1,8) = 9.225$, $p = 0.016$; follow-up t-test, tdTomato, path type, $t(38) = 5.576$, $p < 0.001$; **Figs 1E** and **1F**). In contrast, ChrimsonR-expressing mice continued to frequently choose the high-threat path even after five days of training (ChrimsonR, $F(4,12) = 1.837$, $p = 0.161$; **Fig 1D**) and did not differentiate the running speed between the paths (ChrimsonR, path type, $t(58) = 1.495$, $p = 0.140$; **Figs 1E**, **1F** and S2A). In the random stage (R1-5), both groups gradually reduced the probability of choosing the original low-threat path and no longer showed any differences in the path choice (Session×Group interaction, $F(4,32) = 0.588$, $p = 0.674$; Session, $F(4,32) = 7.747$, $p < 0.001$; Group, $F(1,8) = 2.269$, $p = 0.170$.; **Fig 1D**) or speed (Fig S2A).

Because tdTomato-expressing mice learned to differentiate the speed toward the threat sites depending on threat probabilities (**Fig 1E**), we used the speed variations across laps to gauge the mice's expectation of threats: a slower speed indicated that they expected to receive threats, while

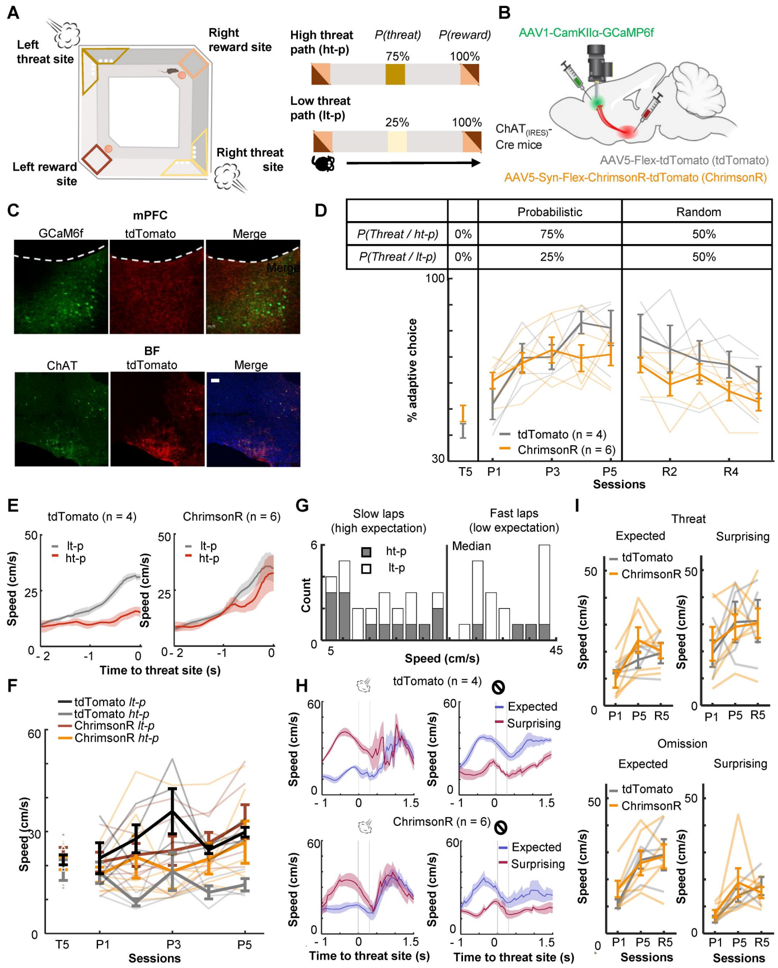

Fig 1.

Optogenetic stimulation of BF-mPFC cholinergic projections during threats impairs discrimination of two probabilistic outcome contingencies.

(**A**) The maze used in the probabilistic spatial learning task (left). The probability of threat and reward assigned to each of two paths connecting two reward sites (right). Mice received air puffs 75% of the time on one of the paths (high-threat path) and 25% of the time on the other path (low-threat path). The color of the squares indicates the locations on the maze as specified in the illustration on the left. (**B**) The site of GRIN lens implantation and viral vector infusion. (**C**) Images showing ChrimsonR-tdTomato expressing cholinergic terminals (red) near the GCaMP6f-expressing neurons (green) in the mPFC (top). Images showing colocalization of ChAT and virally infected cells expressing tdTomato in the BF (bottom). The scale bar represents 100 μm. **(D)** The proportion of laps in which mice chose the low-threat path (% adaptive path choice, one thin line/mouse, mean ± s.e.m.). The table on top indicates the assigned threat probability for each path. Adaptive path choice in the pre-training (T5) and random stage (R1-5) was defined as choosing the path that was the low-threat path during the probabilistic stage (P1-5). (**E**) The movement speed of mice while they ran toward the threat sites in P5 (mean ± s.e.m.). (**F**) The approaching speed toward the threat sites on two paths (one thin line/mouse, mean ± s.e.m.). The speed was averaged over a 500-msec window before the threat site entry. (**G**) The distribution of the speed approaching the threat site in all laps in P5 in a representative mouse. Laps with the approaching speed below or above the median were categorized as “slow” and “fast” laps, respectively. Laps on the high-threat path (grey) were likely to be slow laps, while laps on the low-threat path (white) were likely to be fast laps, consistent with differential threat expectations between the paths. (**H**) The movement speed around the threat sites in fast and slow laps in P5 (mean ± s.e.m.). The speed in fast laps with threats and slow laps with omissions showed the mice’s reaction to surprising outcomes (red). Conversely, the speed in slow laps with threats and fast laps with omissions showed their reaction to expected outcomes (blue). (**I**) The averaged speed during a 500-msec window starting from the threat site entry (one thin line/mouse, mean ± s.e.m.). Laps were categorized into the combination of the approaching speed (fast or slow) and outcomes (threat or omission) and labeled as “expected” and “surprising” as defined in H.

a faster speed implied that they did not (**Fig 1G**). The comparison of the running speed between these two lap types revealed that the mice reacted differently to the same outcomes depending on their expectations. Specifically, mice accelerated more robustly after surprising threats than expected threats (**Fig 1H**, left). They also took longer to accelerate after surprising omissions than expected omissions (**Fig 1H**, right). Similar patterns were also detected in P1 and R5 (Fig S2B). In all these sessions, cholinergic terminal stimulation did not affect the reactions to expected or surprising outcomes (Threat: three-way mixed ANOVA, Session×Expectation×Group interactions, p = 0.285; Session, F(2,16) = 5.626, p = 0.014; Expectation, F(1,8) = 37.918, p < 0.001; Group, F(1,8) = 0.039, p = 0.849; Omission: Session×Expectation×Group interactions, p = 0.360; Session, F(2,16) = 12.840, p < 0.001; Expectation, F(1,8) = 33.280, p < 0.001; Group, F(1,8) = 0.156, p = 0.703; **Fig 1I**). These findings suggest that phasic cholinergic signals do not modulate the development of threat expectation but are essential for evaluating different probabilistic outcome contingencies.

Enhanced phasic cholinergic signals augment the threat-evoked activity of PL cells

To probe neural mechanisms behind the observed behavioral effects, we used a head-mounted single-photon epifluorescence miniature microscope to monitor Ca^{2+} activity of PL cells via a gradient-index (GRIN) lens (**Fig 2A**). The biological and optical crosstalk of optogenetic stimulation on calcium imaging is negligible(68). We first investigated how cholinergic terminal stimulation affected the threat-evoked activity of PL cells by using the data during the first session with probabilistic air-puff delivery with LED stimulation (P1; **Fig 2B**). ~15% of PL cells increased Ca^{2+}activity in response to the combined delivery of air puff and LED stimulation (**Fig 2C**; tdTomato, 14.7%; ChrimsonR, 17.9%). In the absence of LED stimulation, the magnitude of their puff-evoked activity was reduced in ChrimsonR-expressing mice but not in tdTomato-expressing mice (**Fig 2C**). Overall, the LED stimulation increased the puff-evoked activity more robustly in ChrimsonR-expressing than tdTomato-expressing mice (Kolmogorov-Smirnov test, w/ LED, p = 0.016; **Fig 2D**). This group difference was not detected without the LED stimulation (w/o LED, p = 0.386), confirming that the enhanced puff-evoked activity was due to the stimulation of the BF cholinergic terminals.

We also examined the effect of cholinergic terminal stimulation on PL cell activity in the absence of air-puffs. In one pre-training session (T3), LED stimulation was applied when mice entered the corner of the maze, a location that would become the threat site in subsequent days (Fig S3A). Some cells changed their activity at the corner, and the LED stimulation disrupted the location-selective activity. However, the degree of the LED-induced change was comparable between the groups (Kolmogorov-Smirnov test, p = 0.237; Fig S3B). Thus, the optogenetic stimulation barely affected PL cell activity unless it coincided with the natural activation of cholinergic cells by threats(58,59).

Enhanced phasic cholinergic signals suppressed learning-dependent changes in outcome-related PL cell activity

Prefrontal cells are known to change responses to outcomes throughout learning, indicating that they encode values of expected outcomes(69–73) and the discrepancies between expected and actual outcomes (i.e., prediction errors)(41,49–54). Therefore, we next asked how PL cells changed outcome-related activity with learning and how this process was affected by cholinergic terminal stimulation. We categorized laps into four types based on which path the mice took and whether they received an air puff. For laps with air-puff delivery, we only used ones without LED stimulation to isolate the learning-dependent changes from the direct effects of LED stimulation (**Figs 2C** and **2D**). In each lap, the cell activity was aligned to the moment when the mouse's position tracking entered the threat site. In the first (P1) and last (P5) sessions of the probabilistic stage, a sizable proportion of PL cells changed their activity in response to threat

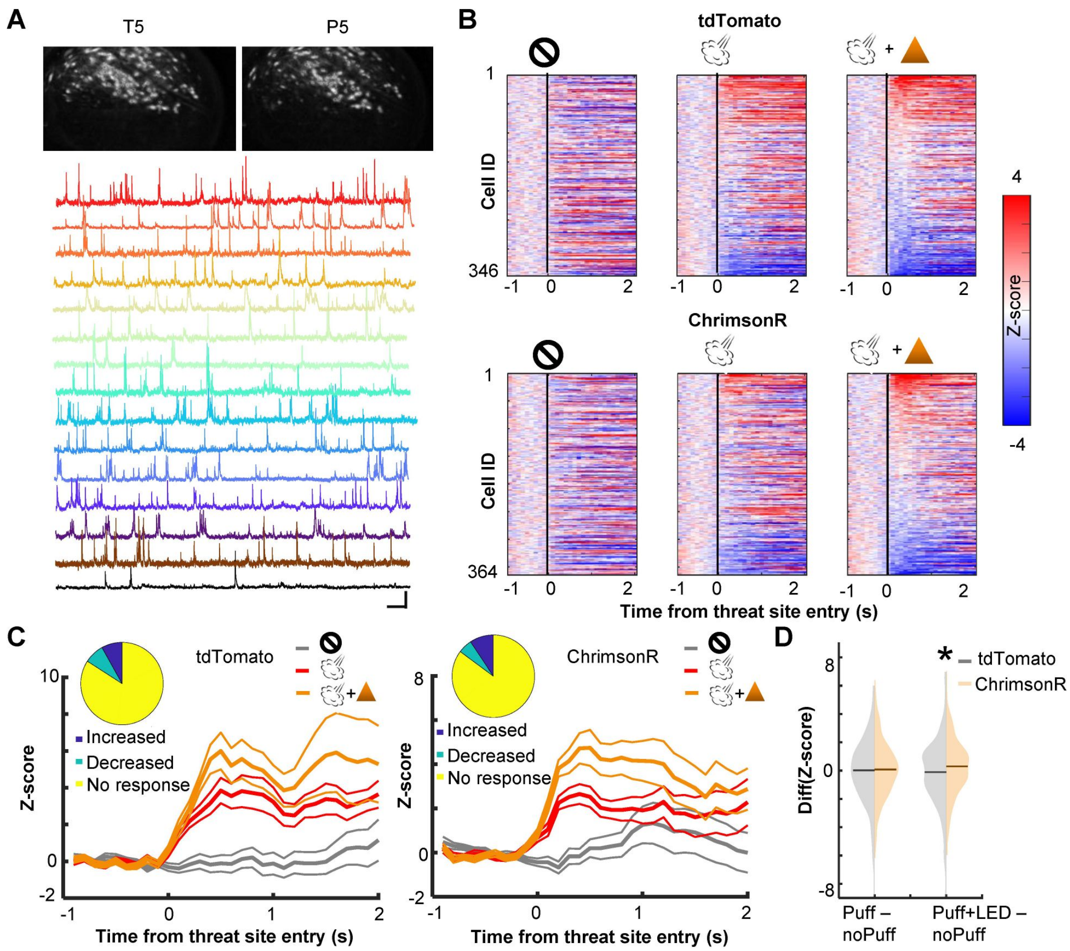

Fig 2.

Cholinergic terminal stimulation augments PL cell activity in response to threats.

(**A**) Field of views in a representative mouse from the last pre-training session (T5) and the fifth probabilistic session (P5; top). White contours represent registered cells of the corresponding session. Calcium traces of extracted cells (bottom). Scale bars represent 50 s × 5 z-score. (**B**) Pseudocolor plots showing the z-normalized cell activity aligned to the threat site entry during puff-omission (left), puff (middle), and puff and LED (right) laps. In all panels, cells were sorted by the response magnitude in puff and LED laps. (**C**) Z-normalized cell activity aligned to the threat site entry of threat-responding cells (mean ± s.e.m.). (inset) The proportion of cells with increased activity (blue) and those with decreased activity (green). (**D**) The distribution of the change in cell activity evoked by air puffs or air puffs and LED. Horizontal lines show the median. Z-normalized activity was averaged over a 500-msec window starting from the threat site entry in each lap type. The average activity in omission laps was subtracted from the average activity in puff laps or puff and LED laps.

delivery or omission (**Figs 3Aa and Ba**). In both groups, cells responding to outcomes on the high-threat path were distinct from those on the low-threat path (P1, Fig S4A; P5, **Fig 3Ab and Bb**), suggesting that cholinergic terminal stimulation did not affect the differentiation of outcome representations between two threat sites. In tdTomato-expressing mice, the omission-evoked activity on the high-threat path was shifted toward excitation from P1 to P5, while the threat-evoked activity was unchanged (Kolmogorov-Smirnov test, threat on ht-p, $p = 0.485$; omission on ht-p, $p = 0.005$; **Fig 3Ac**). Threat- or omission-evoked activity on the low-threat path was not changed across sessions (threat on lt-p, $p = 0.106$; omission on lt-p, $p = 0.718$). In ChrimsonR-expressing mice, the distribution of neither threat-nor omission-evoked responses was changed from P1 to P5 on either path (threat on ht-p, $p = 0.265$; omission on ht-p, $p = 0.055$; threat on lt-p, $p = 0.279$; omission on lt-p, $p = 0.473$; **Fig 3Bc**). The comparisons between the groups revealed that cholinergic terminal stimulation had no effects on threat- or omission-evoked activity in P1 (threat on ht-p, $p = 0.731$; omission on ht-p, $p = 0.682$; threat on lt-p, $p = 0.404$; omission on lt-p, $p = 0.142$; Fig S4B, left); however, it blocked the learning-dependent increase in the responses to rare threats omissions on the high-threat path (omission on ht-p, $p < 0.001$; Fig S4B, right). Thus, enhanced phasic cholinergic signals do not change how PL cells respond to subsequent outcomes within minutes; instead, they modulate the gradual restructuring of outcome representations over days.

Phasic cholinergic signals augmented the selectivity of PL cells for signed prediction errors

The strengthening of cell activity to rare threat omission alluded to the sensitivity of PL cells to the discrepancy between expected and actual outcomes, i.e., prediction errors. To further address this point, we investigated whether PL cells differentiated the outcome-related activity depending on the lap-by-lap fluctuations in threat expectations. Using the speed toward the threat site (**Fig 1G**), laps with air-puff delivery were split into surprising (faster speed) and expected (slower speed) threat laps. In parallel, laps with air-puff omission were split into surprising (slower speed) and expected (faster speed) omission laps. We then quantified the degree to which individual cells differentiated the outcome-related activity depending on expectations by subtracting the activity for expected outcomes from the activity for surprising outcomes. We observed strong differentiations of threat- and omission-evoked activity in all three sessions (**Fig 4A**). In both groups, cells showed diverse patterns of differential activity (**Fig 4B**). Some cells increased the activity in response to surprising threats but decreased the activity in response to surprising threat omission (**Fig 4C**, Up-Down), while others showed reversed patterns (Down-Up). Thus, these cells changed the response direction based on whether outcomes were better or worse than expected, a signature of signed prediction errors. In parallel, other cells showed the same direction of responses to surprising threats and surprising omission (Up-Up, Down-Down), suggesting that they were sensitive to unsigned prediction errors. In parallel, other cells showed expectation-dependent differentiations only to threats or their omissions (Fig S5). In P1, the proportion of cells encoding signed prediction errors was comparable to that encoding unsigned prediction errors (**Fig 4D**, left). These cells were much fewer than others that encode errors of one outcome type. All these cell types were detected in both groups in similar proportions. In P5, however, cholinergic terminal stimulation increased the proportion of cells encoding signed prediction errors (binominal test, $p < 0.05$; **Fig 4D**, middle). As a result, cells that encoded signed prediction errors became twice as common as those that encoded unsigned prediction errors. The overall proportion of cells encoding prediction errors was slightly increased. Similar patterns were also detected during R5 (binominal test, $p < 0.05$; **Fig 4D**, right). These findings suggest that enhanced phasic cholinergic signals augmented the encoding of signed prediction errors in the PL.

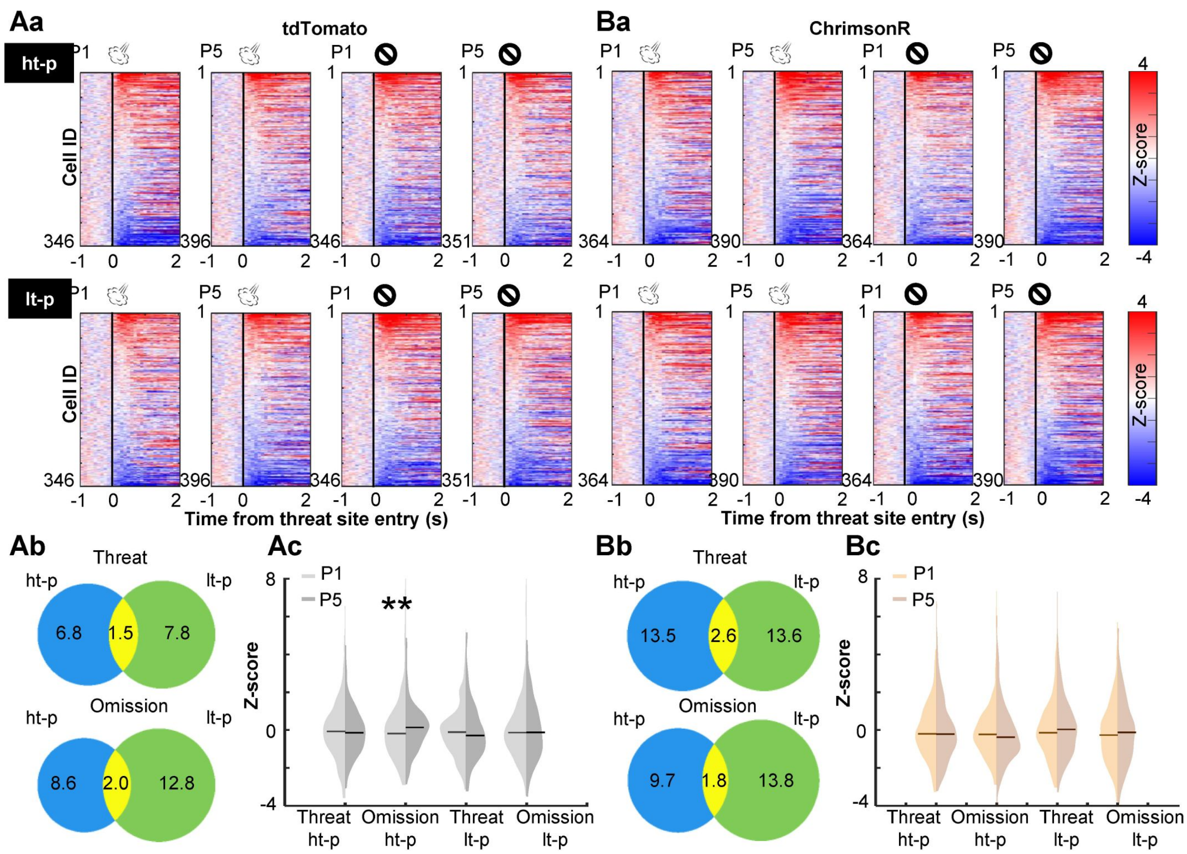

Fig 3.

Cholinergic terminal stimulation abolishes learning-induced strengthening of PL cell responses to rare threat omission.

(**A**) The changes in PL cell activity upon the threat site entry in tdTomato mice. a. Pseudocolor plots showing the z-normalized cell activity aligned to the threat site entry during the first (P1) and last session (P5) of the probabilistic stage. Laps were categorized into four types based on the path (high-threat path, ht-p; low-threat path, lt-p) and the threat state (threat delivery or omission). Cells were sorted by their activity in the threat site in each panel. b. Venn diagrams depicting the proportion of cells responding to threats and their omissions on the ht-p (blue) and the lt-p (green) as well as their overlap (yellow) in P5. The numbers show the percentage of cells in each category. c. The distribution of z-normalized activity in the threat sites in P1 (light grey) and P5 (dark grey) with horizontal bars depicting the median. (**B**) The same as A for PL cell activity in ChrimsonR-expressing mice.

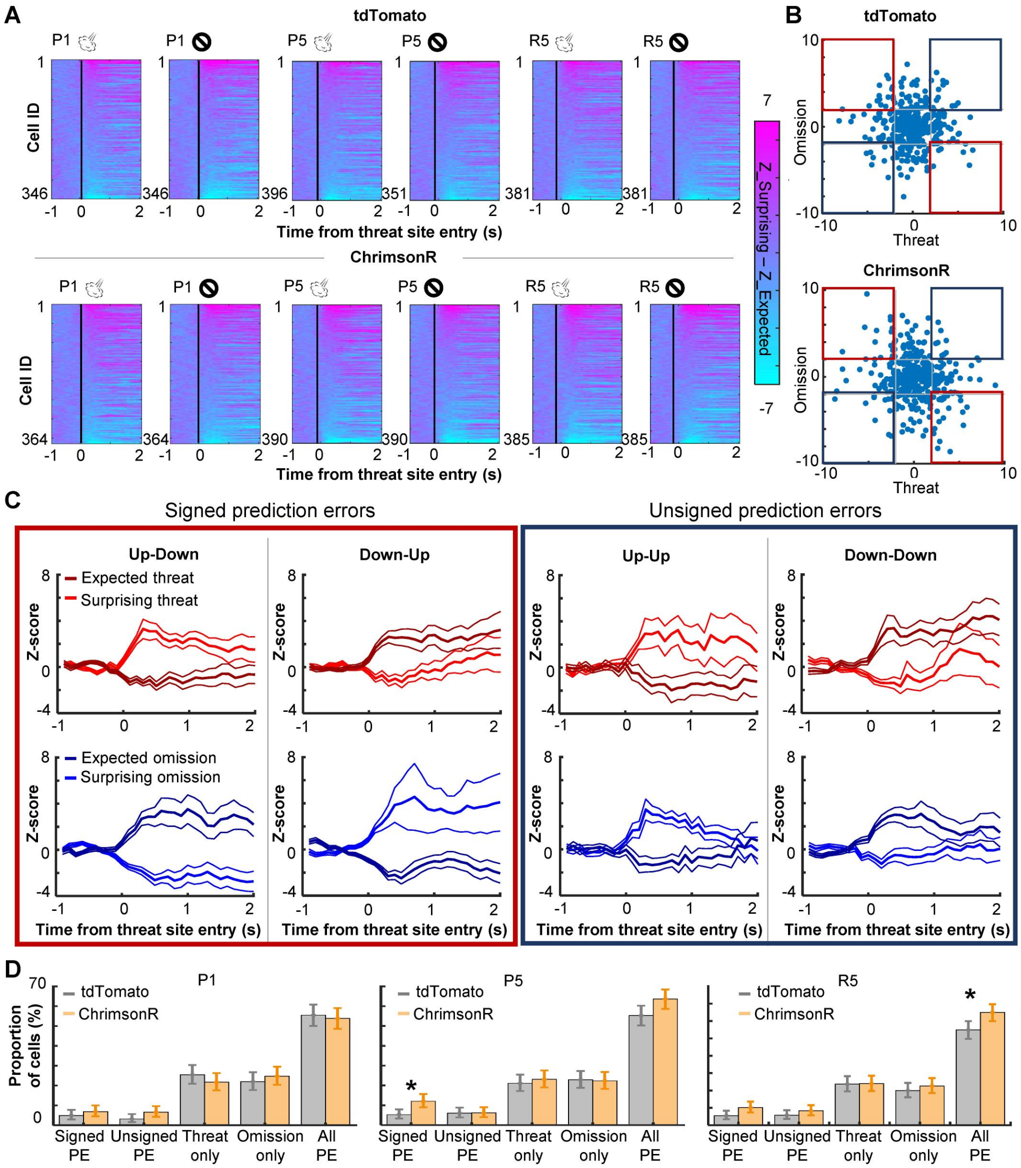

Fig 4.

PL cells differentiate outcome-related activity depending on outcome expectations.

(**A**) Pseudocolor plots showing the differentiation of outcome-related activity between expected and surprising outcomes during the first (P1) and last session (P5) of the probabilistic stage. The color depicts the difference in z-normalized cell activity between laps with expected outcomes and lap with surprising outcomes. Cells were sorted by the activity difference in the threat site in each panel. (**B**) Expectation-dependent differentiation of cell activity in response to threats (x axis) and their omission (y axis; one dot per cell) in P5. The z-normalized activity was averaged during a 500-msec window starting from the threat site entry. The averaged values in laps with expected outcomes were subtracted from those with surprising outcomes (differential activity). Blue squares highlight cells with the same response direction to surprising threats and omissions (unsigned prediction errors). Red squares highlight cells with an opposite response direction to surprising threats and omissions (signed prediction errors). (**C**) The averaged z-normalized activity across cells with different types of activity differentiation (mean ± s.e.m.). A cell was selected when its differential activity for threats or their omission was greater than 2 (Up) or smaller than -2 (Down). Cells with significant differential activity for both threat delivery and omission were then further categorized into four types depending on their response direction. The activity of cells with significant differentiation only for one outcome type was depicted in Fig S5. (**D**) The proportion of cells in various differential activity types. Cells selective for outcome expectations were categorized into four types as defined in C and Fig S5. Error bars show the upper and lower confidence limit (α = 0.05). PE, prediction errors. R5, the last session of the random stage.

Enhanced phasic cholinergic signal impairs adaptive learning of uncertain, but not certain, threats

The analyses of cell activity revealed that enhanced threat-evoked cholinergic signals modified the encoding of prediction errors in the PL. To further investigate how such changes in PL outcome coding affected behavior, we conducted follow-up behavioral experiments with bilateral excitation of the BF cholinergic terminals in the PL. AAVs were infused into the HDB of ChAT-cre mice to express either channelrhodopsin (ChR2, n = 10), ChrimsonR (n = 8) with control fluorophore, or control fluorophore alone (No-opsin, n = 14) in cholinergic cells (Figs S6A, 5A and 5B). Their terminals in the PL (Fig S6B) were excited during every air-puff delivery in all sessions of the probabilistic paradigm [**Fig 5Ca**; *P(Threat|ht-p)* = 75%, *P(Threat|lt-p)* = 25%]. Consistent with the effect of the unilateral manipulation (**Fig 1D**), enhanced threat-evoked cholinergic terminal activity impaired the development of preference for the low-threat path (two-way mixed ANOVA, Session×Group interaction, $F(8,116) = 2.249$, $p = 0.029$; follow-up one-way repeated measure ANOVA, No opsin, $F(4,52) = 11.190$, $p < 0.001$; ChR2, $F(4,36) = 0.627$, $p = 0.646$; ChrimsonR, $F(4,28) = 0.856$, $p = 0.502$; **Figs 5Cb** and 5Cc). Because the task performance of the ChR2 and ChrimsonR groups was comparable, these groups were collapsed into a single group (opsin group) in subsequent analyses. Closer examinations of the movement pattern revealed that mice in the no-opsin group took longer to reach the threat site on the high-than the low-threat path (three-way mixed ANOVA, Path×Group×Session interaction, $F(4,96) = 0.508$, $p = 0.730$; Path×Group interaction, $F(1,24) = 13.54$, $p = 0.001$; follow up t-test, no-opsin, path type, $t(108) = 3.979$, $p < 0.001$; Fig S7Aa). In contrast, mice in the opsin group took comparable time on both paths (path type, $t(148) = 0.570$, $p = 0.569$; Fig S7Aa). In contrast, the time taken from the threat site to the reward site was similar between the groups, and it became shorter across the sessions (Session, $F(4, 96) = 5.091$, $p < 0.001$; Group, $F(1, 24) = 3.846$, $p = 0.062$; Path, $F(1, 24) = 0.297$, $p = 0.591$; Fig S7Ab). Similarly, both groups initially increased and later decreased the time to initiate a subsequent lap (Session, $F(44, 96) = 7.474$, $p < 0.001$; Group, $F(1, 24) = 0.529$, $p = 0.474$; Path, $F(1, 24) = 0.653$, $p = 0.427$; Fig S7Ac).

As outlined in the introduction, the control of learning rate is particularly essential when threats occur unreliably. To test whether cholinergic modulation of the PL is vital for probabilistic threat learning or threat learning in general, we conducted the same experiment with the deterministic paradigm [*P(Threat|ht-p)* = 100%, *P(Threat|lt-p)* = 0%]. We observed that all three groups preferred the path without threats (no-threat path) over the path with threats (threat path; two-way mixed ANOVA, Session×Group interaction, $F(4,24) = 3.195$, $p = 0.031$; follow-up one-way repeated measure ANOVA, no-opsin, $F(2,12) = 25.83$, $p < 0.001$; ChR2, $F(2,6) = 14.39$, $p = 0.005$; ChrimsonR, $F(2,6) = 7.199$, $p = 0.025$; **Fig 5D**). In addition, both groups took longer to reach the threat site on the threat than no-threat path (three-way mixed ANOVA, Session, $F(1, 13) = 0.034$, $p = 0.857$; Group, $F(1, 13) = 0.758$, $p = 0.399$; Path, $F(1, 13) = 26.38$, $p < 0.001$; Fig S7Ba). They also took longer to reach the reward site on the no-threat than threat path (Session, $F(1, 13) = 0.008$, $p = 0.930$; Group, $F(1, 13) = 0.025$, $p = 0.877$; Path, $F(1, 13) = 73.320$, $p < 0.001$; Fig S7Bb). On the contrary, both groups increased the time to initiate a subsequent lap after choosing the no-threat path (Session, $F(1,13) = 5.182$, $p = 0.040$; Group, $F(1, 13) = 0.089$, $p = 0.770$; Path, $F(1, 13) = 48.28$, $p < 0.001$; Fig S7Bc).

Intact learning in the deterministic paradigm (**Fig 5D**) suggests that enhanced phasic cholinergic signals did not make the mice insensitive to threats or incapable of learning threat locations. To identify specific deficits underlying this behavioral change, we investigated how the mice adjusted their path choice after experiencing threats in the probabilistic paradigm. After receiving an air puff on the high-threat path, both groups chose the other path in the subsequent lap starting from the first session (two-way mixed ANOVA, Session, $F(4,56) = 1.054$, $p = 0.388$; Group, $F(1,14) = 0.131$, $p = 0.723$; **Fig 6A**, left). On the other hand, after receiving an air puff

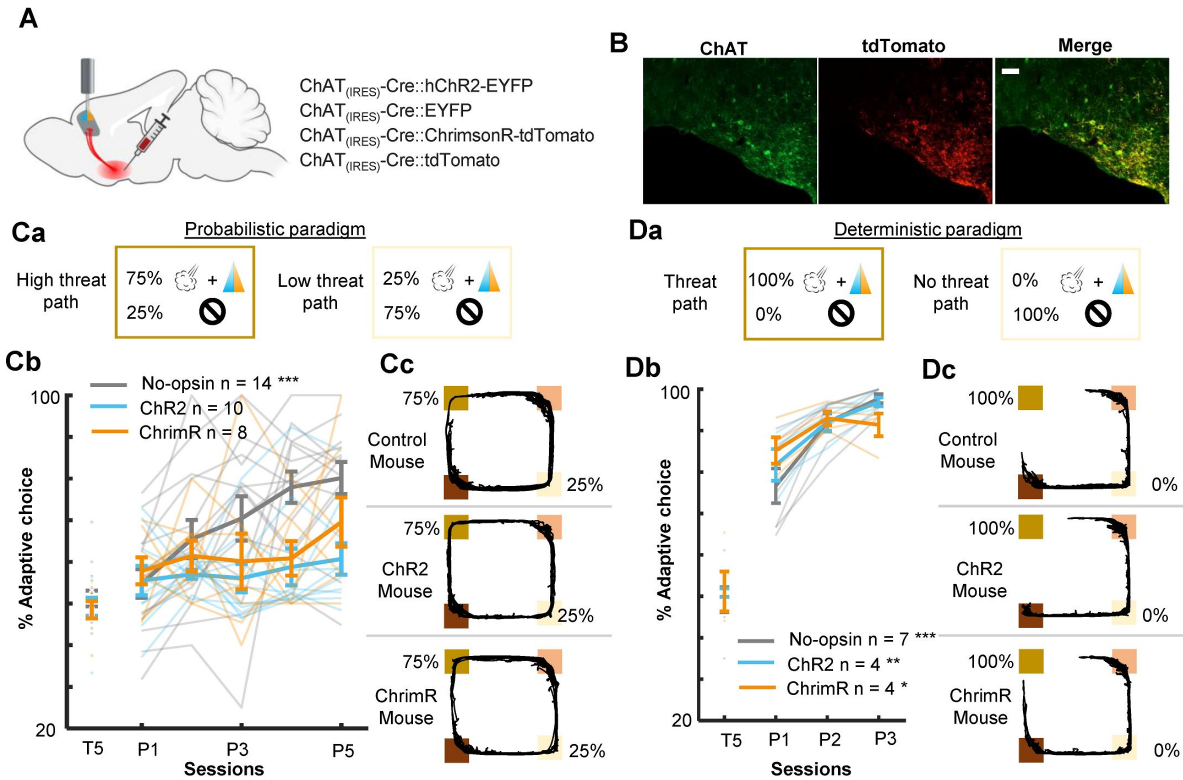

Fig 5.

Threat-evoked phasic cholinergic signals constrain learning from surprising outcomes in a stable but probabilistic environment.

(**A**) Sites of viral vector infusion and optic fiber implantation. (**B**) Representative images showing co-localization of ChAT and virally infected cells expressing tdTomato. The scale bar represents 100 µm. (**C**) The effect of cholinergic terminal stimulation on mice's behavior during the probabilistic paradigm. a. Schematic representation of threat probability and light stimulation on each path. LED stimulation was applied to the PL to activate the BF cholinergic terminals during every threat delivery. b. The percentage of laps in which mice chose the low-threat path. c. Movement trajectories of a representative mouse from each group. The locations of threat and reward sites were specified with the same color scheme as in **Fig 1A**. (**D**) Same as C for the deterministic paradigm.

on the low-threat path, both groups initially chose the other path in the subsequent lap. Over five days of training, however, the no-opsin group learned to stay on the low-threat path, whereas the opsin group continued to switch to the high-threat path (two-way mixed ANOVA, Session×Group interaction, $F(4,44) = 3.442$, $p = 0.017$; follow-up one-way repeated measures ANOVA, No opsin, $F(4,16) = 3.291$, $p = 0.038$; Opsin, $F(4,28) = 0.579$, $p = 0.680$; **Fig 6A**, right).

The above analysis revealed that the no-opsin group learned to ignore rare threats on the low-threat path, but the opsin group did not. This observation led us to theorize that the no-opsin group estimated threat probability by integrating threat occurrences across multiple laps, and this ability might be impaired in the opsin group. To test this idea, we correlated a sequence of path choices against the threat probability estimated with the threat history over different numbers of laps (Fig S7C). Consistent with our view, the path choice of the no-opsin group showed stronger correlations with the estimated threat probability as the greater number of laps was used to calculate threat probability (**Fig 6B**, left). In contrast, the path choice of most mice in the opsin group was not correlated with the estimated threat probability regardless of the number of laps used to calculate the probability (right). The proportion of mice with significant correlations was greater in the no-opsin group than in the opsin group in both paths (chi-square test, ht-p, $X^2_1 = 4.073$, $p = 0.044$; lt-p, $X^2_1 = 7.748$, $p = 0.005$; **Fig 6C**). These observations suggest that abnormally strong cholinergic signals during threats made mice adjust their choices after each threat without tracking its occurrence over time. This would make them highly sensitive to a surprising threat, even when it falls within the expected range of variability.

Discussion

The mPFC has been implicated in the evaluation of probabilistic outcome contingencies (36–39) and contains cells sensitive to outcome expectations(69–73) and prediction errors(41,49–54). Here, we have shown that this function and coding property are modulated by phasic cholinergic signals time-locked to outcomes.

During our probabilistic spatial learning task, PL cells robustly responded to threats and their omission (**Fig 3Aa**). From the beginning, PL cells differentiated outcome-related activity for the two threat sites (Figs S4A and 3Ab), indicating the conjunctive code of outcome and its location. With subsequent learning, omission-evoked activity was strengthened, while threat-evoked activity was unchanged. Notably, such strengthening occurred only for rare threat omissions on the high-threat path but not for frequent omissions on the low-threat path (**Fig 3Ac**), paralleling the gradual differentiation of threat expectations for these paths (**Figs 1D** and **1F**). Furthermore, we also observed that outcome-evoked responses were sensitive to mice's outcome expectations on each lap (**Fig 4A**). Although patterns of activity differentiation were diverse (**Figs 4B**, **4C** and S5), ~10% of PL cells responded to better-than-expected and worse-than-expected outcomes. Half of them reversed the response direction (i.e., increased v.s. decreased activity) for positive and negative errors, indicating the selectivity for signed prediction errors. Meanwhile, the remaining showed the same response direction for both error types, indicating the selectivity for unsigned prediction errors. These findings align with the selectivity of prefrontal cells for reward prediction errors(41,49–54) and expand the critical role of mPFC in prediction error coding to aversive outcomes.

Among these coding properties of PL cells, cholinergic terminal stimulation 1) abolished the learning-dependent increase in the responses to rare threat omission (**Figs 3Bc** and S4B) and 2) increased the proportion of cells encoding signed prediction errors (**Fig 4D**). Notably, these effects emerged over days but not across minutes within the first day, suggesting that enhanced cholinergic signals did not immediately change how PL cells responded to subsequent threats. Instead, it modulated the gradual restructuring of outcome representations with learning. An open question is how phasic cholinergic signals achieve such modulation. Cholinergic modulation of the

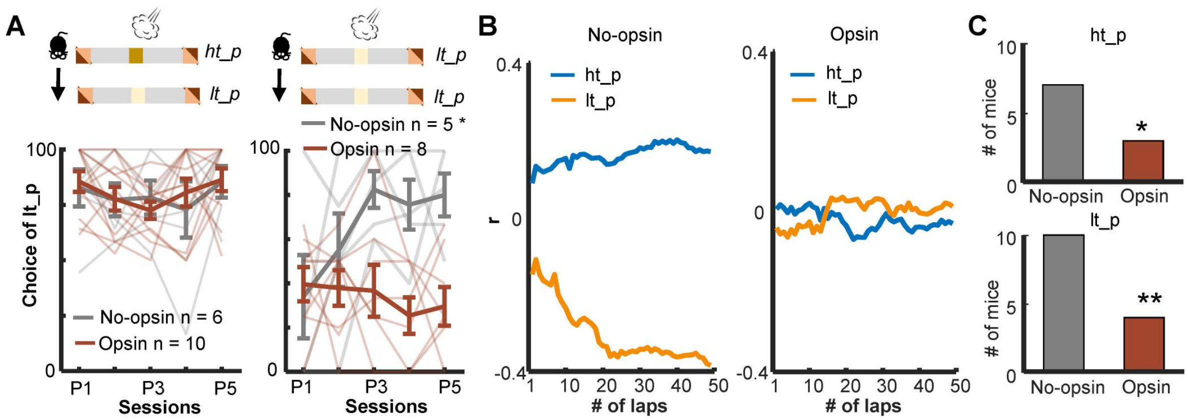

Fig 6.

Enhanced threat-evoked phasic cholinergic signals impair the integration of outcome information over time.

(**A**) The probability of choosing the low-threat path (lt-p) after an air-puff delivery on the high-threat path (ht-p; left; one thin line/mouse, mean ± s.e.m.). Only mice with usable data in all five sessions were used for this analysis. (**B**) Correlation coefficients between the actual path choice and threat probability calculated with the different number of previous laps from representative mice. The greater positive values for the ht-p and negative values for the lt-p indicate stronger correlations with the estimated threat probability (see Fig S7C). (**C**) Numbers of mice showing significant correlations between the estimated threat probability and the choice of the ht-p (top).

mPFC is highly complex because both nicotinic and muscarinic ACh receptors are expressed in pyramidal cells and interneurons in a layer-dependent manner(74,75). Indeed, *in vitro* slice preparations, the application of ACh inhibits pyramidal cells in superficial layers but excites pyramidal cells in deep layers(76–78). These mixed effects at least partially explain why our optogenetic stimulation did not result in a drastic increase in PL cells responding to threats (**Fig 2**) or the corner of the maze (Fig S3).

In parallel to the modulation of cell excitability, ACh also controls the plasticity of glutamatergic synapses(79,80) by modulating glutamate release onto pyramidal cells(81,82) or inhibitory synaptic transmission(83). One potential function of such modulation is to adjust the relative impacts of various long-range inputs on PL cells. For example, some BLA cells encode unsigned prediction errors at the time of outcomes(84,85) and outcome-predictive cues(86). Notably, activating the M1 subtype of muscarinic ACh receptor induces long-term depression of field excitatory post-synaptic potentials in BLA-PL pathways(87). Therefore, enhanced cholinergic signals may reduce the influence of BLA inputs carrying unsigned prediction errors via the activation of M1 receptors, thereby increasing the impact of others carrying signed prediction errors, such as dopamine inputs from the ventral tegmental area (VTA). Although some controversy exists on how dopamine neurons respond to aversiveness(88,89), several studies have reported that dopamine neurons are inhibited by unexpected threats(90,91) and excited by unexpected threat omission(92). In addition, dopaminergic signals at the time of outcome modulate aversive learning(93,94). Investigating such hetero-synaptic plasticity would be a promising future direction to deepen our understanding of intricate cholinergic modulation of prefrontal neural computations.

Alternatively, phasic cholinergic signals may convey signed prediction errors by themselves and directly improve the selectivity of PL cells. However, this is unlikely for several reasons. First, BF cholinergic cells do not differentiate their responses between expected and surprising threats(59,65). Second, cholinergic terminals in the PL were activated by threats but not by surprising threat omission(58) (Fig S8). These observations suggest that phasic cholinergic signals covey the presence of aversive stimuli to PL cells but not prediction errors.

To decipher specific deficits that emerged from the modified prediction error coding in the PL, we conducted additional behavioral experiments (**Fig 5**) and analyzed mice's choices on a lap-by-lap basis (**Fig 6**). We observed that cholinergic terminal stimulation did not affect the likelihood of correcting the choice after threats on the high-threat path (**Fig 6A**), suggesting that enhanced phasic cholinergic signals do not affect the learning from aversive outcomes. This observation explains why the manipulation did not affect learning in the deterministic paradigm, in which mice needed to correct the choice after every threat (**Fig 5D**). The intact learning also eliminates the possibility that cholinergic terminal stimulation affected the sensitivity to threats, the motivation to obtain the reward, the discrimination between the two paths, and the learning of threat locations. In contrast, the enhanced cholinergic signals reduced the likelihood of staying on the low-threat path after a surprising threat (**Fig 6A**). Also, it decoupled the choice from the estimated threat probability based on threat occurrence across multiple laps (**Figs 6B** and **6C**). Theories link the handling of surprising outcomes to the processing of prediction error signals(7,35,95). Specifically, when threats occur reliably, the estimate of threat probability should be updated with each prediction error signal to reduce the magnitude of error signals (**Fig 7**). In contrast, when threats occur probabilistically, the magnitude of prediction error signals massively fluctuates depending on whether a threat occurred or not on each occasion. With such large variability, the estimate of threat probability should not be updated by error signals with unusually large magnitudes. Also, to assess whether an error signal is an anomaly, one must track the absolute magnitude of error signals over time(7,35,95). The present behavioral results are consistent with these theories and identify that threat-evoked phasic cholinergic signals serve as an essential input enabling PL cells to properly handle prediction errors for adaptive threat learning.

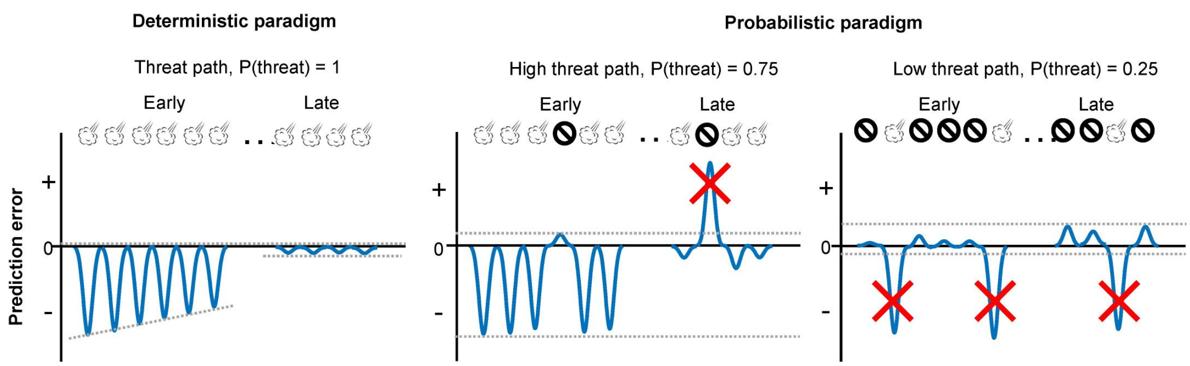

Fig 7

Schematic representations of aversive prediction errors in the deterministic and probabilistic paradigms.

Aversive prediction errors become positive for unexpected threat delivery while negative for unexpected threat omission(97,98). Red crosses indicate prediction error signals with abnormally large amplitude. We argue that the role of outcome-evoked phasic cholinergic signals is to suppress learning from these occasional surprising outcomes.

A question remains open as to whether this function is achieved solely by phasic cholinergic signals within the PL. Although we stimulated cholinergic terminals in the PL, the stimulation could retrogradely activate their cell bodies, leading to additional release of ACh at their terminals in other regions. Indeed, ~75% of PL-projecting cholinergic cells also send axons to the anterior cingulate and orbitofrontal cortex(96). We, therefore, cannot eliminate the possibility that the coordinated release of ACh in multiple prefrontal regions might also contribute to the detected manipulation effects on PL cell activity and adaptive threat learning.

In conclusion, the evidence from cell activity and behavior suggests that the threat-evoked phasic cholinergic signal modulates the gain of learning from surprising threats by controlling the content of outcome coding in the PL. Our findings extend the cholinergic modulation of cognition to an intricate evaluation process of surprising outcomes and its control over adaptive learning and decision-making.

Materials and methods

Ethics statement

All methodological procedures were conducted under the regulation of the University of Toronto Animal Care Committee and the Canadian Council on Animal Care (AUP20012042).

Animals

Adult $ChAT_{(IRES)}$-Cre mice (RRID: IMSR_JAX:028861) were bred as homozygotes. Cre recombinase is expressed under the ChAT promotor in these mice without disrupting endogenous ChAT expression. In total, sixty-eight 8-20 weeks old male $ChAT_{(IRES)}$-Cre mice were used in optogenetics and calcium imaging experiments. Mice were housed under a 12-hour light/dark cycle with *ad libitum* access to food and water. Seven days before the behavioral testing, food was restricted to maintain 85-90% of free-feeding body weight for the experiments.

Viral vectors

AAV9-EF1a-DIO-hChR2(H134R)-EYFP-WPRE-HGHpA (RRID: Addgene_20298), AAV9-EF1a-DIO-EYFP (RRID: Addgene_27056), AAV5-Syn-FLEX-rc[ChrimsonR-tdTomato] (RRID: Addgene_62723) and AAV5-FLEX-tdTomato (RRID: Addgene_28306) were purchased from the Addgene. AAV1-CamKIIa-GCaM6f-WPRE-bGHpA ready-to-image virus (Catalog ID: 1000-002613) was purchased from the Inscopix (Bruker).

Surgery

Viral infusion

Eight to 12 weeks old $ChAT_{(IRES)}$-Cre mice were anesthetized (1-2% isoflurane by volume in oxygen at a flow rate of 0.8 L/min; Fresenius Kabi) and administered with meloxicam (20 mg/kg) subcutaneously. Mice were placed in a stereotaxic frame, and holes were made in the skull bilaterally above the basal forebrain (BF; +0.2 mm AP, ±1.3 mm ML)(99). A 33 G microinfusion cannula was connected to a 10-µl Hamilton syringe with Tygon tubing. One of the viral vectors (AAV9-EF1a-DIO-hChR2(H134R)-EYFP-WPRE-HGHpA, AAV9-EF1a-DIO-EYFP, AAV5-Syn-FLEX-rc[ChrimsonR-tdTomato] and AAV5-FLEX-tdTomato; 0.8-1.0 µl/side) was infused to the basal forebrain (BF; -5.4 mm DV) at a rate of 0.1 µl/min for optogenetic experiments. For imaging experiments, AAV5-Syn-FLEX-rc[ChrimsonR-tdTomato] or AAV5-FLEX-tdTomato (0.8-1 µl/side) was infused to the BF (-5.4 mm DV) at a rate of 0.1 µl/min. Another hole was drilled above the right medial prefrontal cortex (mPFC; +1.9 mm AP, + 0.35 mm ML). A 200 nl AAV1-CamKIIa-GCaM6f-WPRE-bGHpA was infused to the prelimbic region (PL; -2.2 mm DV) at a rate of 50 nl/min. After infusion, the cannula was left in the brain for 15 minutes, allowing the viral vectors to diffuse.

Optic fiber implantation

Three to 6 weeks after viral infusion, mice were placed in a stereotaxic frame as described above. Optic fibers (200 μm core diameter, 0.39 NA; Thorlabs) were implanted bilaterally at 30 degrees lateral to the medial angle above the PL (+1.9 mm AP, ±1.4 mm ML, -1.8 mm DV). The optic fibers were secured to the skull with resin cement (3M dental, Relyx unicem). The mice were given 14 days to recover from the surgery before behavioral testing.

Gradient index (GRIN) lens implantation

Three to 6 weeks after viral infusion, mice were placed in a stereotaxic frame as described above. A 1.2 mm diameter craniotomy was made above the mPFC (centered at +1.9 mm AP, + 0.6 mm ML). A 25 G blunt needle was slowly lowered above the PL (+1.9 mm AP, + 0.6 mm ML, -1.5 mm DV) at a rate of 10 μm/s to make a track for GRIN lens implant. Five minutes later, the 25 G blunt needle was slowly removed at a rate of 10 μm/s. A ProView™ integrated lens (1 mm diameter, 4 mm length; Inscopix) with baseplate attached was implanted 100-200 nm above the PL (+1.9 mm AP, + 0.55 mm ML, -1.7-1.9 mm DV; lowered at a rate of 2 μm/s). To prevent dust, the lens and basal plate were secured to the skull with resin cement (3M dental, Relyx unicem) and covered by a magnetic cover (Inscopix). The mice were given at least 21 days to recover from the surgery before behavioral testing.

Behavioral experiments

Probabilistic spatial learning

The apparatus was a 75 cm × 75 cm square-shaped maze with an 8 cm wide track and 15 cm walls. The inner wall was inclined 30 degrees outward to prevent cables from tangling at the corners. A camera was positioned centrally 1 m above the maze to monitor the mice's movement. At the two opposite corners of the maze, a circle feeding trough (2 cm in diameter) was placed at the rear wall (reward site). A sugar pellet (20 mg, Bio-Serve) was manually delivered to the trough using a Tygon tube accessible through the opening (1 cm diameter) at the rear wall right above the trough. Additionally, two pieces of Tygon tubing (6 mm internal diameter and total length of 20 cm) were attached outside the two sides walls of the other two corners (each side 10 cm long, threat site). The tubes and side walls were perforated with 20 holes × 5 mm diameter. Using plastic Y connectors, these tubes were connected to solenoids and a compressed air cylinder (Whisper-tech). An air puff (300 ms duration, 35 psi) was delivered via this Tygon tubing. During the task, the mouse's head, body, and tail locations were extracted from a video recorded at 30 frames per second by an overhead webcam (Ethovision software, Noldus Information Technology). The entry to the reward or threat sites were defined as the time when the mouse body center entered the pre-defined region of interest (reward site, 20 cm × 20 cm; threat sites, 8 cm × 20 cm), and the corresponding timestamps were stored. In a subset of laps, an air puff was delivered 100 ms after the entry to a threat site, at which Ethovision software emitted a TTL pulse to activate a solenoid valve connected to the tubes installed at that threat site.

During the first five days (pre-training stage), all mice ran on the maze to collect sugar pellets at the reward sites without any air-puff delivery. Subsequently, each mouse underwent one of three paradigms. In the probabilistic paradigm, air puffs were delivered 75% of the time at one threat site and 25% of the time at the other threat site. In the deterministic paradigm, air puffs were delivered 100% of the time at one threat site and 0% of the time at the other threat site. In the random paradigm, air puffs were delivered 50% of the time at both threat sites. Sugar pellets were always delivered in all stages regardless of path choice and air puff deliveries.

1-photon single-cell calcium imaging with optogenetic manipulations

Calcium imaging videos were recorded using a head-mounted miniature microscope (miniscope, Inscopix nVoke acquisition system v2.0) and accompanying software (IDAS, v1.7.1). The fluorescence power during imaging sessions ranged from 0.5-0.8 mW/mm^2 at a rate of 20 Hz, with an excitation wavelength of 455 ± 8 nm. Ethovision software was used to trigger and control the Inscopix Data acquisition (DAQ) box in synchrony with behavioral video tracking (30 fps) using a webcam installed near the ceiling.

Before imaging experiments, mice were handled by the experimenter for seven days. The mice were further habituated to the miniscope mounting procedure by attaching a dummy scope for 2-3 sessions (~10 minutes long). During these sessions, focus depth was adjusted for each animal and was maintained consistently throughout the subsequent imaging sessions. During the same period, these mice were subjected to food restriction to maintain 85-90% of their free-feeding body weight.

The mice underwent the pre-training (5 days), probabilistic (5 days), and random (5 days) stages. To mitigate photobleaching, we only recorded images on a subset of sessions (Fig S1D). A dust cover was removed, and the implanted lens was cleaned with 75% ethanol and double-distilled water before attaching the miniscope to the mouse in each session. The mouse was placed in a cylinder (20 cm diameter, 25 cm height) in the maze's center for 5 minutes to acclimate to the environment. Subsequently, the cylinder was removed, and the mouse was transferred to one of the reward sites, initiating the session automatically. Each imaging session lasted 40 minutes and comprised two LED-ON and two LED-OFF epochs. The order of epochs was counterbalanced across mice and sessions (ON-OFF-ON-OFF or OFF-ON-OFF-ON). During the LED-ON epoch, Arduino sent TTL signals to turn on solenoids to deliver air puffs when the mouse entered a threat site. Concurrently, another TTL signal activated an orange LED integrated into the miniscope for 300 ms (20Hz, 8 mW/mm^2, 590-650 nm wavelength). In the LED-OFF epoch, the orange LED was not activated concurrently with air puffs. In a subset of sessions (P2, P4, R2, and R4), the LED stimulation was applied during every air-puff delivery without capturing calcium imaging videos.

Optogenetic experiments

A separate cohort of mice first underwent the pre-training stage and moved on to the probabilistic or deterministic stage, during which they received bilateral optogenetic manipulations of cholinergic terminals in the PL. Daily sessions lasted 15-20 minutes or paused when mice ran over 60 laps. Each lap started when a mouse exited one reward site and ended when it entered the other reward site. A blue laser (300 ms, 20 Hz, 3-5 mW, 473 nm wavelength; Laserglow Technologies) was turned on during every air-puff delivery for mice expressing ChR2. An orange LED (300 ms, 20 Hz, 5 mW, 625 nm wavelength; M625F2, Thorlabs) was turned on during every air-puff delivery for mice expressing ChrimsonR. These light stimulation parameters were chosen based on previous works(68,94). Although we did not validate the release of ACh in the PL following the optogenetic stimulation, previous works have reported that ACh is released following optogenetic excitation of cholinergic terminals with light stimulation parameters comparable to ours (100,101).

Tissue collection, immunostaining, and image acquisition

Three to 5 days after the last session, mice were subcutaneously administered with avertin (20 mg/kg) and transcardially perfused with 0.9% saline followed by chilled 4% paraformaldehyde (PFA). The brains were immersed in 4% PFA for 48-72 hours, transferred to 30% sucrose/PBS, and stored at 4°C. A series of 50 µm coronal sections were collected from the area containing the mPFC (AP: +1.3 to +2.4 mm) and BF (AP: +0.6 to -0.6 mm) using a cryostat (CM3050S, Leica Biosystems). The tissues were washed in PBS three times and incubated with 10% donkey (# D9663, Sigma-Aldrich, RRID: AB_2810235) or goat serum (# ab156046, Abcam) for 2 hours at room temperature (RT) followed by a three-time wash in PBS.

To check the transgene expression, the BF sections were incubated with primary antibodies against choline acetyltransferase (ChAT; Goat, # AB144P, 1:200, Millipore, RRID: AB_90661) for 24-36 hours at 4 °C. Following three times wash in PBS, the brain sections were then incubated with a secondary antibody (Rhodamine-conjugated donkey-anti-goat, # 705-025-147, 1:200, Jackson ImmunoResearch, RRID: AB_2340389) for 2 hours at RT. Finally, the BF sections were mounted using an antifade mountant with DAPI (# P36935, ThermoFisher Scientific). Images for viral spread and optic fiber track identification were collected using a confocal microscope (Zeiss) under a 10× objective.

Behavioral analysis

X- and y-coordinates of a mouse's position were extracted from the video tracking data. The timestamps were collected when a mouse entered/exited the threat and reward sites. A lap was counted as a "full" lap when a mouse exited a reward site, passed a threat site, and entered the other reward site. A lap was counted as a "half" lap when a mouse exited a reward site, entered a threat site, and returned to the same reward site. If a mouse exited a reward site and returned to the same reward site without entering a threat site, it was not counted as a half lap. Adaptive path choice was defined as the percentage of the laps in which a mouse chose the path associated with a lower threat probability among all full and half laps.

The speed approaching the threat site was calculated as the average speed in a 500-msec window immediately before the threat site entry during each lap. In each session, the median of the approaching speed was used to categorized laps into fast and slow laps (**Fig 1G**). The reaction to outcomes was quantified as the average speed in a 500-msec window immediately after the threat site entry.

Latency to threats was defined as the time difference between the reward site exit and the threat site entry. Latency to reward was defined as the time difference between the threat site exit and the reward site entry. The latency to initiate a subsequent lap was defined as the time in a reward site before initiating a new lap.

To examine if threat presence in a previous lap affected path choice in the subsequent lap, we used the data in the probabilistic stage, in which air puffs were delivered 75% of the time on one path (high-threat path) and 25% of the time on the other path (low-threat path). We first selected laps with air-puff delivery on each path in each session. If the number of these laps was less than two in a session, that session was discarded. We then selected mice with usable data in all five sessions. For each path, we calculated the proportion of laps in which a mouse chose the low-threat path in the subsequent lap.

To investigate how many laps mice used to estimate threat probabilities associated with the paths, we examined the relationship between a lap-by-lap path choice and the threat probability estimated based on the history of threats over different numbers of laps. In each mouse, laps were concatenated across all five sessions of the probabilistic stage. We then coded path choice in each lap as 1 for the high-threat path and 0 for the low-threat path (Fig S7C). We also calculated threat probability based on threat occurrence in the past 1-50 laps. The Pearson correlation coefficient was calculated between the path choice and the estimated threat probability. We then counted the mice that showed a significant correlation with threat probability calculated with any lap number.

PL imaging data analysis

Cell extraction

All imaging data were imported into the Inscopix Data Processing Software (IDPS) to spatially down-sample by a factor of 2 and temporally down-sample by a factor of 2. Pre-processed movies underwent spatial bandpass filer (low cut-off = 0.005 pixel-1, high cut-off = 0.500 pixel-1) and motion correction (mean frame) using IDPS. We used an established cell-detection algorithm, constrained nonnegative matrix factorization (CNMFe) in motion-corrected movies for cell detection. The applied CNMFe setting was minCorr = 0.8, minPeak2Noise = 10, Merging Threshold = 0.7. Each extracted region of interest (ROI) was manually checked to select the potential neuronal traces from the noise.

Outcome-evoked responses

In each cell, Ca^{2+} activity in 3-second windows starting from 1 second before the threat site entry was collected and binned into 100-millisecond bins. The binned activity was averaged across laps and converted into z-scores with the mean and standard deviation (SD) of activity during a 1-second window before the threat site entry (first ten bins). This normalization procedure was applied to each lap type separately. Specifically, to test the effect of LED stimulation on threat-evoked activity (**Fig 2**), laps were categorized into three types: air-puff omission, air-puff delivery, and air-puff delivery with LED stimulation. To examine how PL cells changed threat- and omission-evoked responses with learning (**Fig 3**), laps were categorized into four types: by which path a mouse took and whether it received air puff or not. To isolate learning-dependent changes in puff-evoked responses from the direct effect of LED stimulation, laps with air-puff delivery and LED stimulation were not included in this analysis. Lastly, to investigate how PL cells differentiated outcome-evoked responses depending on outcome expectation (**Fig 4**), laps were categorized into four types based on how fast a mouse approached the threat site and whether it received air puff or not.

The response magnitude was calculated by averaging z-normalized activity over a 500-msec window starting from the threat site entry. Cells were deemed to respond to threats or their omission (**Figs 3Ab** and Bb) if the response magnitude was greater than 2. Cells were deemed to differentiate outcome-evoked responses depending on outcome expectation (**Fig 4C**) if the difference in the response magnitude between surprising and expected outcomes was greater than 2 or smaller than -2.

Statistical analysis

The sample size was chosen based on previous studies using similar experimental approaches(102–104) . The data were presented as the group mean ± standard error of the mean unless otherwise specified. Statistical analyses were performed with GraphPad Prism, JASP (*https://jasp-stats.org/*), or MATLAB. To determine the statistical significance, we used one-way (repeated) analysis of variance (ANOVA), two-way mixed/repeated-measures ANOVA, unpaired t-test, chi-square test, Kolmogorov-Smirnov test, and binominal test. Degrees of freedom were corrected with Greenhouse-Geisser correction when sphericity was violated. For *post hoc* analyses, Tukey's multiple comparison test was used following one-way repeated measures ANOVA. Significance was defined as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Acknowledgements

This work was supported by NSERC Discovery Grant, CFI Leaders Opportunity Fund (Kaori Takehara-Nishiuchi), and NSERC graduate fellowship (Gaqi Tu).

Author Contributions

Conceptualization: Kaori Takehara-Nishiuchi, Gaqi Tu.

Data curation: Gaqi Tu, Peiying Wen, Adel Halawa.

Formal analysis: Kaori Takehara-Nishiuchi, Gaqi Tu.

Funding acquisition: Kaori Takehara-Nishiuchi, Gaqi Tu.

Investigation: Gaqi Tu, Kaori Takehara-Nishiuchi.

Methodology: Gaqi Tu, Peiying Wen, Adel Halawa.

Project administration: Gaqi Tu.

Supervision: Kaori Takehara-Nishiuchi.

Visualization: Gaqi Tu, Kaori Takehara-Nishiuchi.

Writing – original draft: Kaori Takehara-Nishiuchi.

Writing – review & editing: Gaqi Tu, Kaori Takehara-Nishiuchi.

References

1. Rescorla R, Wagner AR., Black AB, Prokasy WF (1972) **A theory of Pavlovian conditioning : Variations in the effectiveness of reinforcement and nonreinforcement** *Classical conditioning: current research and theory* Appleton-Century-Crofts

2. Sutton RS (1988) **Learning to predict by the methods of temporal differences** *Mach Learn* **3**:9–44

3. Mackintosh NJ (1975) **A theory of attention: Variations in the associability of stimuli with reinforcement** *Psychol Rev* **82**:276–98

4. Pearce JM, Hall G (1980) **A model for Pavlovian learning: Variations in the effectiveness of conditioned but not of unconditioned stimuli** *Psychol Rev* **87**:532–52

5. Yu AJ, Dayan P (2005) **Uncertainty, neuromodulation, and attention** *Neuron* **46**:681–92

6. Bach DR, Dolan RJ (2012) **Knowing how much you don't know: a neural organization of uncertainty estimates** *Nat Rev Neurosci* **13**:572–86

7. Soltani A, Izquierdo A (2019) **Adaptive learning under expected and unexpected uncertainty** *Nat Rev Neurosci* **20**:635–44

8. Clarke HF, Dalley JW, Crofts HS, Robbins TW, Roberts AC (2004) **Cognitive inflexibility after prefrontal serotonin depletion** *Science* **304**:878–80

9. Boulougouris V, Glennon JC, Robbins TW (2008) **Dissociable effects of selective 5-HT2A and 5-HT2C receptor antagonists on serial spatial reversal learning in rats** *Neuropsychopharmacol Off Publ Am Coll Neuropsychopharmacol* **33**:2007–19

10. Brigman JL, Mathur P, Harvey-White J, Izquierdo A, Saksida LM, Bussey TJ, et al. (2010) **Pharmacological or genetic inactivation of the serotonin transporter improves reversal learning in mice** *Cereb Cortex N Y N* **20**:1955–63

11. Matias S, Lottem E, Dugué GP, Mainen ZF (2017) **Activity patterns of serotonin neurons underlying cognitive flexibility** *eLife* **6**

12. Iigaya K, Fonseca MS, Murakami M, Mainen ZF, Dayan P (2018) **An effect of serotonergic stimulation on learning rates for rewards apparent after long intertrial intervals** *Nat Commun* **9**

13. Grossman CD, Bari BA, Cohen JY (2022) **Serotonin neurons modulate learning rate through uncertainty** *Curr Biol CB* **32**:586–599

14. Rowe JB, Saunders JR, Durantou F, Robbins TW (1996) **Systemic idazoxan impairs performance in a non-reversal shift test: implications for the role of the central noradrenergic systems in selective attention** *J Psychopharmacol Oxf Engl* **10**:188–94

15. Tait DS, Brown VJ, Farovik A, Theobald DE, Dalley JW, Robbins TW (2007) **Lesions of the dorsal noradrenergic bundle impair attentional set-shifting in the rat** *Eur J Neurosci* **25**:3719–24

16. Alexander JK, Hillier A, Smith RM, Tivarus ME, Beversdorf DQ (2007) **Beta-adrenergic modulation of cognitive flexibility during stress** *J Cogn Neurosci* **19**:468–78

17. Mueller D, Porter JT, Quirk GJ (2008) **Noradrenergic signaling in infralimbic cortex increases cell excitability and strengthens memory for fear extinction** *J Neurosci Off J Soc Neurosci* **28**:369–75

18. Uematsu A, Tan BZ, Ycu EA, Cuevas JS, Koivumaa J, Junyent F, et al. (2017) **Modular organization of the brainstem noradrenaline system coordinates opposing learning states** *Nat Neurosci* **20**:1602–11

19. Lapiz MDS, Morilak DA (2006) **Noradrenergic modulation of cognitive function in rat medial prefrontal cortex as measured by attentional set shifting capability** *Neuroscience* **137**:1039–49

20. Marshall L, Mathys C, Ruge D, de Berker AO, Dayan P, Stephan KE, et al. (2016) **Pharmacological Fingerprints of Contextual Uncertainty** *PLoS Biol* **14**

21. Fitzgerald PJ, Giustino TF, Seemann JR, Maren S (2015) **Noradrenergic blockade stabilizes prefrontal activity and enables fear extinction under stress** *Proc Natl Acad Sci U S A* **112**:E3729–3737

22. Floresco SB, Magyar O, Ghods-Sharifi S, Vexelman C, Tse MTL (2006) **Multiple dopamine receptor subtypes in the medial prefrontal cortex of the rat regulate set-shifting** *Neuropsychopharmacol Off Publ Am Coll Neuropsychopharmacol* **31**:297–309

23. Stopper CM, Tse MTL, Montes DR, Wiedman CR, Floresco SB (2014) **Overriding phasic dopamine signals redirects action selection during risk/reward decision making** *Neuron* **84**:177–89

24. Jenni NL, Larkin JD, Floresco SB (2017) **Prefrontal Dopamine D1 and D2 Receptors Regulate Dissociable Aspects of Decision Making via Distinct Ventral Striatal and Amygdalar Circuits** *J Neurosci Off J Soc Neurosci* **37**:6200–13

25. Onge JR St, Abhari H, Floresco SB (2011) **Dissociable contributions by prefrontal D1 and D2 receptors to risk-based decision making** *J Neurosci Off J Soc Neurosci* **31**:8625–33

26. Ragozzino ME (2002) **The effects of dopamine D(1) receptor blockade in the prelimbic-infralimbic areas on behavioral flexibility** *Learn Mem Cold Spring Harb N* **9**:18–28

27. Witte EA, Davidson MC, Marrocco RT (1997) **Effects of altering brain cholinergic activity on covert orienting of attention: comparison of monkey and human performance** *Psychopharmacology (Berl)* **132**:324–34

28. Chiba AA, Bushnell PJ, Oshiro WM, Gallagher M (1999) **Selective removal of cholinergic neurons in the basal forebrain alters cued target detection** *Neuroreport* **10**:3119–23

29. Phillips JM, McAlonan K, Robb WG, Brown VJ (2000) **Cholinergic neurotransmission influences covert orientation of visuospatial attention in the rat** *Psychopharmacology (Berl)* **150**:112–6

30. Vossel S, Bauer M, Mathys C, Adams RA, Dolan RJ, Stephan KE, et al. (2014) **Cholinergic stimulation enhances Bayesian belief updating in the deployment of spatial attention** *J Neurosci Off J Soc Neurosci* **34**:15735–42

31. Hassani SA, Neumann A, Russell J, Jones CK, Womelsdorf T (2023) **M1-selective muscarinic allosteric modulation enhances cognitive flexibility and effective salience in nonhuman primates** *Proc Natl Acad Sci U S A* **120**

32. Rushworth MFS, Behrens TEJ (2008) **Choice, uncertainty and value in prefrontal and cingulate cortex** *Nat Neurosci* **11**:389–97

33. Shackman AJ, Salomons TV, Slagter HA, Fox AS, Winter JJ, Davidson RJ (2011) **The integration of negative affect, pain and cognitive control in the cingulate cortex** *Nat Rev Neurosci* **12**:154–67

34. Behrens TEJ, Woolrich MW, Walton ME, Rushworth MFS (2007) **Learning the value of information in an uncertain world** *Nat Neurosci* **10**:1214–21

35. Soltani A, Koechlin E (2022) **Computational models of adaptive behavior and prefrontal cortex** *Neuropsychopharmacol Off Publ Am Coll Neuropsychopharmacol* **47**:58–71

36. Onge JR St, Floresco SB (2010) **Prefrontal cortical contribution to risk-based decision making** *Cereb Cortex N Y N* **20**:1816–28

37. Paine TA, O'Hara A, Plaut B, Lowes DC (2015) **Effects of disrupting medial prefrontal cortex GABA transmission on decision-making in a rodent gambling task** *Psychopharmacology (Berl)* **232**:1755–65

38. Zeeb FD, Baarendse PJJ, Vanderschuren LJMJ, Winstanley CA (2015) **Inactivation of the prelimbic or infralimbic cortex impairs decision-making in the rat gambling task** *Psychopharmacology (Berl)* **232**:4481–91

39. Orsini CA, Heshmati SC, Garman TS, Wall SC, Bizon JL, Setlow B (2018) **Contributions of medial prefrontal cortex to decision making involving risk of punishment** *Neuropharmacology* **139**:205–16

40. Matsumoto K, Suzuki W, Tanaka K (2003) **Neuronal correlates of goal-based motor selection in the prefrontal cortex** *Science* **301**:229–32

41. Amiez C, Joseph JP, Procyk E (2006) **Reward encoding in the monkey anterior cingulate cortex** *Cereb Cortex N Y N* **16**:1040–55

42. Takehara-Nishiuchi K, McNaughton BL (2008) **Spontaneous changes of neocortical code for associative memory during consolidation** *Science* **322**:960–3

43. Yan R, Wang T, Ma X, Zhang X, Zheng R, Zhou Q (2021) **Prefrontal inhibition drives formation and dynamic expression of probabilistic Pavlovian fear conditioning** *Cell Rep* **36**

44. Jezzini A, Bromberg-Martin ES, Trambaiolli LR, Haber SN, Monosov IE (2021) **A prefrontal network integrates preferences for advance information about uncertain rewards and punishments** *Neuron* **109**:2339–2352

45. Jacobs DS, Moghaddam B (2020) **Prefrontal Cortex Representation of Learning of Punishment Probability During Reward-Motivated Actions** *J Neurosci* **40**:5063–77

46. Passecker J, Mikus N, Malagon-Vina H, Anner P, Dimidschstein J, Fishell G, et al. (2019) **Activity of Prefrontal Neurons Predict Future Choices during Gambling** *Neuron* **101**:152–164

47. Park J, Moghaddam B (2017) **Risk of punishment influences discrete and coordinated encoding of reward-guided actions by prefrontal cortex and VTA neurons** *eLife* **6**

48. Choi K, Piasini E, Díaz-Hernández E, Cifuentes LV, Henderson NT, Holly EN, et al. (2023) **Distributed processing for value-based choice by prelimbic circuits targeting anterior-posterior dorsal striatal subregions in male mice** *Nat Commun* **14**

49. Bryden DW, Johnson EE, Tobia SC, Kashtelyan V, Roesch MR (2011) **Attention for learning signals in anterior cingulate cortex** *J Neurosci Off J Soc Neurosci* **31**:18266–74

50. Hayden BY, Heilbronner SR, Pearson JM, Platt ML (2011) **Surprise signals in anterior cingulate cortex: neuronal encoding of unsigned reward prediction errors driving adjustment in behavior** *J Neurosci Off J Soc Neurosci* **31**:4178–87

51. Matsumoto M, Matsumoto K, Abe H, Tanaka K (2007) **Medial prefrontal cell activity signaling prediction errors of action values** *Nat Neurosci* **10**:647–56

52. Seo H, Lee D (2007) **Temporal filtering of reward signals in the dorsal anterior cingulate cortex during a mixed-strategy game** *J Neurosci Off J Soc Neurosci* **27**:8366–77

53. Hyman JM, Holroyd CB, Seamans JK (2017) **A Novel Neural Prediction Error Found in Anterior Cingulate Cortex Ensembles** *Neuron* **95**:447–456

54. Kehrer P, Brigman JL, Cavanagh JF (2023) **Depth recordings of the mouse homologue of the Reward Positivity** *Cogn Affect Behav Neurosci*

55. Yu AJ, Dayan P (2002) **Acetylcholine in cortical inference** *Neural Netw Off J Int Neural Netw Soc* **15**:719–30

56. Bloem B, Poorthuis RB, Mansvelder HD (2014) **Cholinergic modulation of the medial prefrontal cortex: the role of nicotinic receptors in attention and regulation of neuronal activity** *Front Neural Circuits* **8**

57. Gielow MR, Zaborszky L (2017) **The Input-Output Relationship of the Cholinergic Basal Forebrain** *Cell Rep* **18**:1817–30

58. Tu G, Halawa A, Yu X, Gillman S, Takehara-Nishiuchi K (2022) **Outcome-Locked Cholinergic Signaling Suppresses Prefrontal Encoding of Stimulus Associations** *J Neurosci Off J Soc Neurosci*

59. Hangya B, Ranade SP, Lorenc M, Kepecs A (2015) **Central Cholinergic Neurons Are Rapidly Recruited by Reinforcement Feedback** *Cell* **162**:1155–68

60. Harrison TC, Pinto L, Brock JR, Dan Y (2016) **Calcium Imaging of Basal Forebrain Activity during Innate and Learned Behaviors** *Front Neural Circuits* **10**

61. Ruivo LM Teles-Grilo, Baker KL, Conway MW, Kinsley PJ, Gilmour G, Phillips KG, et al. (2017) **Coordinated Acetylcholine Release in Prefrontal Cortex and Hippocampus Is Associated with Arousal and Reward on Distinct Timescales** *Cell Rep* **18**:905–17

62. Guo W, Robert B, Polley DB (2019) **The Cholinergic Basal Forebrain Links Auditory Stimuli with Delayed Reinforcement to Support Learning** *Neuron* **103**:1164–1177

63. Laszlovszky T, Schlingloff D, Hegedüs P, Freund TF, Gulyás A, Kepecs A, et al. (2020) **Distinct synchronization, cortical coupling and behavioral function of two basal forebrain cholinergic neuron types** *Nat Neurosci* **23**:992–1003

64. Robert B, Kimchi EY, Watanabe Y, Chakoma T, Jing M, Li Y, et al. (2021) **A functional topography within the cholinergic basal forebrain for encoding sensory cues and behavioral reinforcement outcomes** *eLife* **10**

65. Hegedüs P, Sviatkó K, Király B, Martínez-Bellver S, Hangya B (2023) **Cholinergic activity reflects reward expectations and predicts behavioral responses** *iScience* **26**

66. Mineur YS, Picciotto MR (2021) **The role of acetylcholine in negative encoding bias: too much of a good thing?** *Eur J Neurosci* **53**:114–25

67. Cools R, Arnsten AFT (2022) **Neuromodulation of prefrontal cortex cognitive function in primates: the powerful roles of monoamines and acetylcholine** *Neuropsychopharmacol Off Publ Am Coll Neuropsychopharmacol* **47**:309–28

68. Stamatakis AM, Schachter MJ, Gulati S, Zitelli KT, Malanowski S, Tajik A, et al. (2018) **Simultaneous Optogenetics and Cellular Resolution Calcium Imaging During Active Behavior Using a Miniaturized Microscope** *Front Neurosci* **12**

69. Schoenbaum G, Chiba AA, Gallagher M (1998) **Orbitofrontal cortex and basolateral amygdala encode expected outcomes during learning** *Nat Neurosci* **1**:155–9

70. Wallis JD, Miller EK (2003) **Neuronal activity in primate dorsolateral and orbital prefrontal cortex during performance of a reward preference task** *Eur J Neurosci* **18**:2069–81

71. Padoa-Schioppa C, Assad JA (2006) **Neurons in the orbitofrontal cortex encode economic value** *Nature* **441**:223–6

72. Kim S, Hwang J, Lee D (2008) **Prefrontal coding of temporally discounted values during intertemporal choice** *Neuron* **59**:161–72

73. Sul JH, Kim H, Huh N, Lee D, Jung MW (2010) **Distinct roles of rodent orbitofrontal and medial prefrontal cortex in decision making** *Neuron* **66**:449–60

74. Picciotto MR, Higley MJ, Mineur YS (2012) **Acetylcholine as a neuromodulator: cholinergic signaling shapes nervous system function and behavior** *Neuron* **76**:116–29

75. Venkatesan S, Lambe EK (2020) **Chrna5 is essential for a rapid and protected response to optogenetic release of endogenous acetylcholine in prefrontal cortex** *J Neurosci* **40**:7255–68

76. Kassam SM, Herman PM, Goodfellow NM, Alves NC, Lambe EK (2008) **Developmental Excitation of Corticothalamic Neurons by Nicotinic Acetylcholine Receptors** *J Neurosci* **28**:8756–64

77. Gulledge AT, Bucci DJ, Zhang SS, Matsui M, Yeh HH (2009) **M1 receptors mediate cholinergic modulation of excitability in neocortical pyramidal neurons** *J Neurosci Off J Soc Neurosci* **29**:9888–902

78. Poorthuis RB, Bloem B, Schak B, Wester J, de Kock CPJ, Mansvelder HD (2013) **Layer-Specific Modulation of the Prefrontal Cortex by Nicotinic Acetylcholine Receptors** *Cereb Cortex N Y NY* **23**:148–61

79. Verhoog MB, Obermayer J, Kortleven CA, Wilbers R, Wester J, Baayen JC, et al. (2016) **Layer-specific cholinergic control of human and mouse cortical synaptic plasticity** *Nat Commun* **7**

80. Sabec MH, Wonnacott S, Warburton EC, Bashir ZI (2018) **Nicotinic Acetylcholine Receptors Control Encoding and Retrieval of Associative Recognition Memory through Plasticity in the Medial Prefrontal Cortex** *Cell Rep* **22**:3409–15

81. Lambe EK, Picciotto MR, Aghajanian GK (2003) **Nicotine induces glutamate release from thalamocortical terminals in prefrontal cortex** *Neuropsychopharmacol Off Publ Am Coll Neuropsychopharmacol* **28**:216–25

82. Wang BW, Liao WN, Chang CT, Wang SJ (2006) **Facilitation of glutamate release by nicotine involves the activation of a Ca2+/calmodulin signaling pathway in rat prefrontal cortex nerve terminals** *Synap N Y N* **59**:491–501

83. Couey JJ, Meredith RM, Spijker S, Poorthuis RB, Smit AB, Brussaard AB, et al. (2007) **Distributed network actions by nicotine increase the threshold for spike-timing-dependent plasticity in prefrontal cortex** *Neuron* **54**:73–87

84. Belova MA, Paton JJ, Morrison SE, Salzman CD (2007) **Expectation modulates neural responses to pleasant and aversive stimuli in primate amygdala** *Neuron* **55**:970–84

85. Roesch MR, Calu DJ, Esber GR, Schoenbaum G (2010) **Neural correlates of variations in event processing during learning in basolateral amygdala** *J Neurosci Off J Soc Neurosci* **30**:2464–71

86. Klavir O, Genud-Gabai R, Paz R (2013) **Functional connectivity between amygdala and cingulate cortex for adaptive aversive learning** *Neuron* **80**:1290–300

87. Maksymetz J, Joffe ME, Moran SP, Stansley BJ, Li B, Temple K, et al. (2019) **M1 Muscarinic Receptors Modulate Fear-Related Inputs to the Prefrontal Cortex: Implications for Novel Treatments of Posttraumatic Stress Disorder** *Biol Psychiatry* **85**:989–1000

88. Mirenowicz J, Schultz W (1996) **Preferential activation of midbrain dopamine neurons by appetitive rather than aversive stimuli** *Nature* **379**:449–51

89. Fiorillo CD (2013) **Two dimensions of value: dopamine neurons represent reward but not aversiveness** *Science* **341**:546–9

90. Matsumoto M, Hikosaka O (2009) **Two types of dopamine neuron distinctly convey positive and negative motivational signals** *Nature* **459**:837–41

91. Matsumoto H, Tian J, Uchida N, Watabe-Uchida M (2016) **Midbrain dopamine neurons signal aversion in a reward-context-dependent manner** *eLife* **5**

92. Salinas-Hernández XI, Vogel P, Betz S, Kalisch R, Sigurdsson T, Duvarci S (2018) **Dopamine neurons drive fear extinction learning by signaling the omission of expected aversive outcomes** *eLife* **7**

93. Luo R, Uematsu A, Weitemier A, Aquili L, Koivumaa J, McHugh TJ, et al. (2018) **A dopaminergic switch for fear to safety transitions** *Nat Commun* **9**

94. Weele CM Vander, Siciliano CA, Matthews GA, Namburi P, Izadmehr EM, Espinel IC, et al. (2018) **Dopamine enhances signal-to-noise ratio in cortical-brainstem encoding of aversive stimuli** *Nature* **563**:397–401

95. Preuschoff K, Bossaerts P (2007) **Adding prediction risk to the theory of reward learning** *Ann N Y Acad Sci* **1104**:135–46

96. Chandler D, Waterhouse BD (2012) **Evidence for broad versus segregated projections from cholinergic and noradrenergic nuclei to functionally and anatomically discrete subregions of prefrontal cortex** *Front Behav Neurosci* **6**

97. McNally GP, Johansen JP, Blair HT (2011) **Placing prediction into the fear circuit** *Trends Neurosci* **34**:283–92

98. Iordanova MD, Yau JOY, McDannald MA, Corbit LH (2021) **Neural substrates of appetitive and aversive prediction error** *Neurosci Biobehav Rev* **123**:337–51

99. Paxinos G., Franklin K.J. (2019) **Paxinos and Franklin's the Mouse Brain in Stereotaxic Coordinates** Academic Press

100. Kimchi EY, Burgos-Robles A, Matthews GA, Chakoma T, Patarino M, Weddington JC, et al. (2024) **Reward contingency gates selective cholinergic suppression of amygdala neurons** *eLife* **12**

101. Gritton HJ, Howe WM, Mallory CS, Hetrick VL, Berke JD, Sarter M (2016) **Cortical cholinergic signaling controls the detection of cues** *Proc Natl Acad Sci* **113**:E1089–97

102. Cho KKA, Shi J, Phensy AJ, Turner ML, Sohal VS (2023) **Long-range inhibition synchronizes and updates prefrontal task activity** *Nature* **617**:548–54

103. Kin K, Francis-Oliveira J, Kano SI, Niwa M (2023) **Adolescent stress impairs postpartum social behavior via anterior insula-prelimbic pathway in mice** *Nat Commun* **14**

104. Sun C, Yang W, Martin J, Tonegawa S (2020) **Hippocampal neurons represent events as transferable units of experience** *Nat Neurosci* **23**:651–63

Author information

Gaqi Tu

Department of Psychology, University of Toronto, Toronto, Canada, Collaborative Program in Neuroscience, University of Toronto, Toronto, Canada
ORCID iD: 0000-0001-5807-0798

Peiying Wen

Department of Psychology, University of Toronto, Toronto, Canada

Adel Halawa

Human Biology Program, University of Toronto, Toronto, Canada

Kaori Takehara-Nishiuchi
Department of Psychology, University of Toronto, Toronto, Canada, Collaborative Program in Neuroscience, University of Toronto, Toronto, Canada, Department of Cell and Systems Biology, University of Toronto, Toronto, Canada
ORCID iD: 0000-0002-7282-7838

For correspondence: kaori.nishiuchi@utoronto.ca

Editors

Reviewing Editor
Alicia Izquierdo
University of California, Los Angeles, Los Angeles, United States of America

Senior Editor
Kate Wassum
University of California, Los Angeles, Los Angeles, United States of America

Reviewer #1 (Public review):

Tu, Wen, et al. investigated the activity of mPFC putative glutamatergic neurons during a probabilistic threat discrimination and avoidance learning task using miniaturized GRIN lens implantation and single-photon calcium imaging in freely moving mice. In conjunction with this cellular recording, they employed channelrhodopsin-mediated optogenetic excitation of terminals from basal forebrain cholinergic projection neurons coupled to the delivery of an air puff on either of two maze paths with differential threat probability. The authors found that the optogenetic manipulation altered mPFC encoding of outcomes and disrupted animals' behavioral adaptation. Over the course of multiple learning sessions, optogenetically stimulated mice lagged behind control animals in resolving the differential threat probabilities on the two paths and making adaptive choices. In particular, the animals with optogenetic stimulation of cholinergic terminals were significantly more likely to switch to the path with higher threat probability after having just gotten a rare air puff on the generally "safer" path. Combined with data from a deterministic version of the task showing that optogenetically stimulated mice could behaviorally discriminate between the paths appropriately under such circumstances, these results suggest an impairment in the experimental animals' ability to make use of threat history over multiple trials. This comparison of probabilistic and deterministic versions of the same task is a highlight of this paper, representing a thoughtfulness about what information can be gleaned from such variations in the design of behavioral experiments that is all too often lacking. These data are timely in contributing to an ongoing discussion in the field about the role of phasic cholinergic signaling to the cortex, about which relatively little is known.

While the ensemble recording of mPFC neurons during the task appears to be reliable and well-designed and the behavioral effects of the optogenetic stimulation are convincing, some major weaknesses of the paper limit its usefulness to others in the field:

(1) Optogenetic excitation of presynaptic terminals can lead to antidromic action potentials that alter the firing properties of the target cell (see the excellent review on challenges of and strategies for presynaptic optogenetic experiments Rost et al., Nat Neurosci 2022). To their credit, the authors explicitly acknowledge this fact, but they believe that the only alternative possibility is that their intervention could lead to increased acetylcholine release at collateral projections in other prefrontal subregions. In fact, we do not know that the mechanism mediating the behavioral changes observed involves acetylcholine at all, as many ChAT+

basal forebrain neurons co-transmit using GABA (Saunders et al., Nature, 2015; Saunders et al., eLife, 2015; Granger et al., Neuropharmacology, 2016). A very useful internal control, which is recommended by Rost et al. for such presynaptic excitation experiments, would be to locally infuse nicotinic or muscarinic cholinergic antagonists into the mPFC in an attempt to reverse the optogenetically induced deficit; this would resolve whether the effect is indeed mediated by cholinergic neurotransmission and if it is specific to the mPFC.

(2) In a similar vein, the fact that LED illumination in the no-opsin control group appears to increase activity in prefrontal neurons (Figure 2C) and, moreover, has a functional effect in disrupting location-selective cellular activity to a similar extent as in the ChrimsonR group (Figure S3) is inadequately explained and cause for concern. Although the authors argue that the degree or "robustness" of puff-evoked activity was significantly greater in the ChrimsonR group as compared to fluorophore-only controls, their statistical test for demonstrating this is the Kolmogorov-Smirnov test (Figure 2D), thus showing that the two samples likely are drawn from different distributions but little else.

(3) Throughout the paper, the authors rely heavily on the Kolmogorov-Smirnov and binomial tests (Figures 2D, 3, 4D, S3, S4) to compare distributions in this manner, but it is unclear to me why these would be the most appropriate statistical tests for what they seek to demonstrate. Given the holistic nature of these tests in comparing the shape and spread of distributions, I am concerned that they might be inflating the significance of the differences between groups. Even if the authors were seeking a nonparametric statistical test, which most likely would be quite appropriate, there are nonparametric versions of ANOVA that they could use (e.g. Kruskal-Wallis, Friedman). Indeed, in much of this data set a repeated measures statistical analysis would seem to be called for, whereas the Kolmogorov-Smirnov test assumes that the two samples must be independent of each other. The most notable example of this premise being violated is in Figure 3, where data from the same cell populations in the same animals are being compared between experimental days and across various trial types.

https://doi.org/10.7554/eLife.102986.1.sa3

Reviewer #2 (Public review):

Summary:

The authors tested:

(1) Whether mice learn that they are more/less likely to receive an aversive air puff outcome at different corners of a square-shaped open field apparatus, under 75%/25% probabilistic contingencies;

(2) Whether stimulating basal forebrain cholinergic neurons and terminals in the prefrontal cortex affects learning in this context; and

(3) Whether stimulating cholinergic neurons affects prefrontal cortical single neuron calcium signaling about outcome expectations during learning and contingency changes. They found that mice that received cholinergic stimulation approached high and low aversive outcome probability sites at similar velocities, while control mice approached high probability sites slower, suggesting that cholinergic stimulation impaired learning. Cholinergic stimulation reduced cortical neuron calcium activity during trials on the high-probability corner when the outcome was not delivered. The authors provide additional characterization of cellular responses during delivery/omission trials in high/low probability corners, using running speed as a proxy for low versus high expectations. The study will likely be of interest to those who are interested in prediction and error signaling in the cortex; however, the task and analyses do not permit very easy or clear dissociation of prediction versus prediction error

signaling and place field versus place field-expectation multiplexing. The study has several strengths but some weaknesses, which are discussed below.

Strengths:

It is clear the authors were very careful and did a great job with their image processing and segmentation procedures. The details in the methods are appreciated, as are the supplemental descriptive statistics on cell counts.

There are careful experimental controls - for example, the authors showed that the effects of cholinergic stimulation with air puff present are greater than without it, thus ruling out effects of stimulation on cellular physiology that were independent of learning or the task.

The addition of a channelrhodopsin stimulation group is helpful to show that the effects are robust and not wavelength/opsin-specific.

The prefrontal cortex cholinergic terminal stimulation experiment is a great addition. It shows that the behavioral effects of cell body stimulation, which was used in the imaging experiments, are similar to cortical terminal stimulation, where the imaging was performed.

Weaknesses:

The analyses were a bit difficult to follow and therefore it is difficult to determine whether the cells are signaling predictions versus prediction errors - a very important distinction.

The task does not fully dissociate place field coding, since learning about the different probabilities necessarily took place at different areas in the apparatus. Some additional analyses could help address this.

https://doi.org/10.7554/eLife.102986.1.sa2

Reviewer #3 (Public review):

Summary:

Using a combination of optogenetic tools and single-photon calcium imaging, the authors collected a set of high-quality data and conducted thorough analyses to demonstrate the importance of cholinergic input to the prelimbic cortex in probabilistic spatial learning, particularly pertaining to threat.

Strengths:

Given the importance of the findings, this paper will appeal to a broad audience in the systems, behavioural, and cognitive neuroscience community.

Weaknesses:

I have only a few concerns that I consider need to be addressed.

(1) Can the authors describe the basic effect of cholinergic stimulation on PL neurons' activity, during pretraining, probabilistic, and random stages? From the plot, it seems that some neurons had an increase and others had a decrease in activity. What are the percentages for significant changes in activities, given the intensity of stimulation? Were these changes correlated with the neurons' selectivity for the location? If they happen to have the data, a dose-response plot would be very helpful too.

(2) Figure 2B: The current sorting does not show the effects of puff and LED well. Perhaps it's best to sort based on the 'puff with no stim' condition in the middle, by the total activity in 2s

following the puff, and then by the timing in the rise/drop of activity (from early to late). This way perhaps the optogenetic stimulation would appear more striking. Figure 3Aa and Ba have the same issue: by the current sorting, the effects are not very visible at all. Perhaps they want to consider not showing the cells that did not show the effect of puff and/or LED.

Also, I would recommend that the authors use ABCD to refer to figure panels, instead of Aa, Ab, etc. This is very hard to follow.

(3) The authors mentioned the laminar distribution of ACh receptors in discussion. Can they show the presence/absence of topographic distribution of neurons responding to puff and/or LED?

(4) Figure 2C seems to show only neurons with increased activity to an air puff. It's also important to know how neurons with an inhibitory response to air-puff behaved, especially given that in tdTomato animals, the proportion of these neurons was the same as excitatory responders.

(5) Page 5, lines 107 and 110: Following 2-way ANOVA, the authors used a 'follow-up 1-way rmANOVA' and 'follow-up t-test' instead of post hoc tests (e.g. Tukey's). This doesn't seem right. Please use post hoc tests instead to avoid the problem of multiple comparisons.

(6) Figure 1H: in the running speed analysis, were all trials included, both LED+ and LED-? This doesn't affect the previous panels in Figure 1 but it could affect 1H. Did stimulation affect how the running speed recovers?

On a related note, does a surprising puff/omission affect the running speed on the subsequent trial?

(7) On Page 7, line 143, it says "In the absence of LED stimulation, the magnitude of their puff-evoked activity was reduced in ChrimsonR-expressing mice...", but then on line 147 it says "This group difference was not detected without the LED stimulation". I don't follow what is meant by the latter statement, it seems to be conflicting with line 143. The red curves in the left vs right panels do not seem different. The effect of air puff seems to differ, but is this due to a higher gray curve ('no puff' condition) in the ChrimsonR group?

(8) Did the neural activity correlate with running speed? Since the main finding was the absence of difference in running speed modulation by probability in ChrimsonR mice, one would expect to see PL cells showing parallel differences.

https://doi.org/10.7554/eLife.102986.1.sa1

Author response:

> *(1) We do not know that the mechanism mediating the behavioral changes observed involves acetylcholine at all. (Reviewer 1)*

The reviewer rightly pointed out the co-release of acetylcholine (ACh) and GABA from cholinergic terminals. We believe that the detected behavioral changes are because of the augmentation of this innate mixed chemical signal. We agree that identifying the receptor specificity is an essential next step; however, addressing this point requires a currently unavailable research tool to block cholinergic receptors for a few hundred milliseconds. This temporal specificity is vital because acetylcholine is released in the medial prefrontal cortex (mPFC) on two distinct timescales, the slow release over tens of minutes from the task onset and the fast release time-locked to salient stimuli (TelesGrilo Ruivo et al., 2017). Moreover, the former slow signal is far more robust than the latter phasic signal. The pharmacological experiments suggested by the reviewer will suppress both the tonic and phasic signals, making it difficult to interpret the results. Given the rapid technological advancement in this field, we hope to investigate the underlying mechanisms in detail in the future.

> *(2) It is unclear whether mPFC cells are signaling predictions versus prediction errors. (Reviewer 2)*

As the reviewer pointed out, mPFC cells signal the prediction of imminent outcomes (Baeg et al., 2001; Mulder et al., 2003; Takehara-Nishiuchi and McNaughton, 2008; Kyriazi et al., 2020).

However, the key difference between prediction signals and prediction error signals is their time course. The prediction signals begin to arise before the actual outcome occurs, whereas the prediction error signals are emitted after subjects experience the presence or absence of the expected outcome. In all our analyses, cell activity was normalized by the activity during the 1-second window before the threat site entry (i.e., the reveal of actual outcome; Lines 655-659). Also, all the statistical comparisons were made on the normalized activity during the 500-msec window, starting from the threat site entry (Lines 669670). Because this approach isolated the change in cell activity after the actual outcome, we interpret the data in Figure 4C as prediction error signals.

> *(3) The task does not fully dissociate place field coding. (Reviewer 2)*

The present analysis included several strategies to dissociate outcome selectivity from location selectivity (Figure 4). First, we collapsed cell activity on two threat sites to suppress the difference in cell activity between the sites. Second, our analysis compared how cell activity at the same location differed depending on whether outcomes were expected or surprising (Figure 4C). Nevertheless, we can use the present data to investigate the spatial tuning of mPFC cells. Indeed, an earlier version of this manuscript included some characterizations of spatial tuning. However, these data were deemed irrelevant and distracting when this manuscript was reviewed for publication in a different journal. As such, these data were removed from the current version. We are in the process of publishing another paper focusing on the spatial tuning of mPFC cells and their learning-dependent changes.

> *(4) The basic effects of cholinergic terminal stimulation on mPFC cell activity are unclear. (Reviewers 1, 3)*

We acknowledge the lack of characterization of the optogenetic manipulation of cholinergic terminals on mPFC cell activity outside the task context. As outlined in the discussion section (Lines 309-321), cholinergic modulation of mPFC cell activity is highly complex and most likely varies depending on behavioral states. In addition, because we intended to augment naturally occurring threatevoked cholinergic terminal responses (Tu et al., 2022), our optogenetic stimulation parameters were 3-5 times weaker than those used to evoke behavioral changes solely by the optogenetic stimulation of cholinergic terminals (Gritton et al., 2016). Based on these points, we validated the optogenetic stimulation based on its effects on air-puff-evoked cell activity during the task (Figure 2C, 2D).

(5) Some choices of statistical analyses are questionable (Reviewers 1, 3)

We used the Kolmogorov-Smirnov (KS) test to investigate whether the distribution of cell responses differed between the two groups (Figure 2D) or changed with learning (Figure 3Ac, 3Bc). As seen in Figure 3Aa, some mPFC cells increased calcium activity in response to air-puffs, while others decreased. We expected that the manipulation or learning would alter these responses. If they are strengthened, the increased responses will become more positive, while the decreased responses will become more negative. If they are weakened, both responses will become closer to 0. Under such conditions, the shape of the distribution of cell response will change but not the median. The KS test can detect this, but not other tests sensitive to the difference in medians, such as Wilcoxon rank-sum tests. In Figure 2D, KS tests were applied to the independently sampled data from the control and ChrimsonRexpressing mice. In Figure 3Ac and 3Bc, we used all cells imaged in the first and fifth sessions. Considering that ~50% of them were longitudinally registered on both days, we acknowledge the violation in the assumption of independent sampling. In Figure 1D, we detected significant interaction between the group and sessions. Several approaches are appropriate to demonstrate the source of this interaction. We chose to conduct one-way ANOVA separately in each group to demonstrate the significant change in % adaptive choice across the sessions in the control group but not the ChrimsonR group. The cutoff for significance was adjusted with the Bonferroni correction in follow-up paired t-tests used in Figure 1F.

https://doi.org/10.7554/eLife.102986.1.sa0